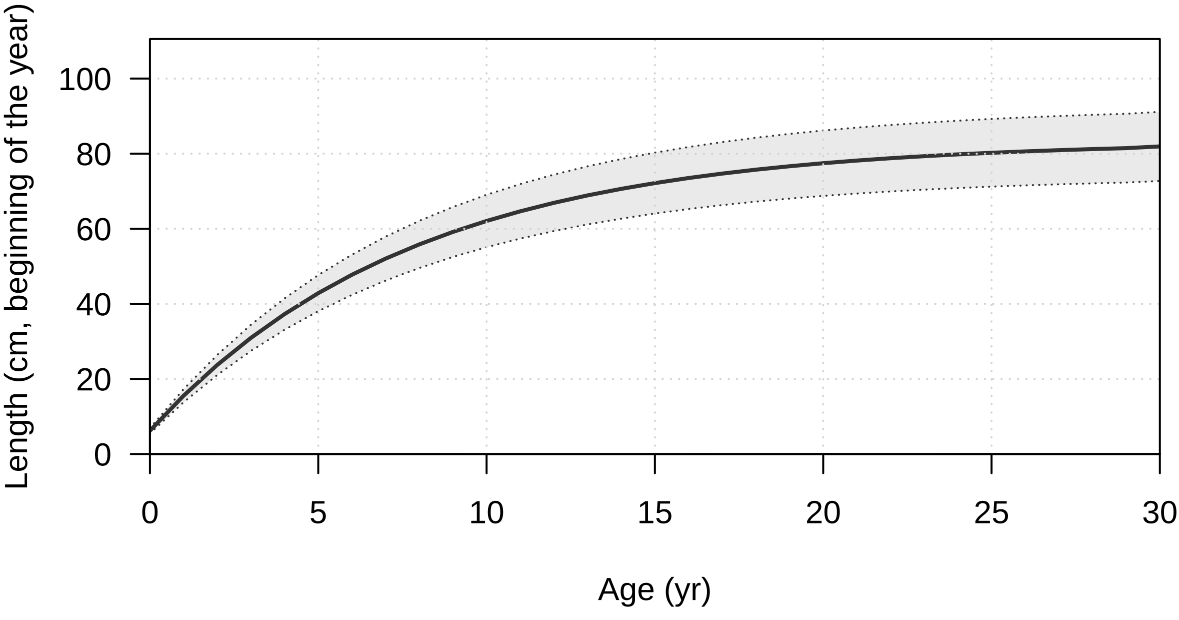
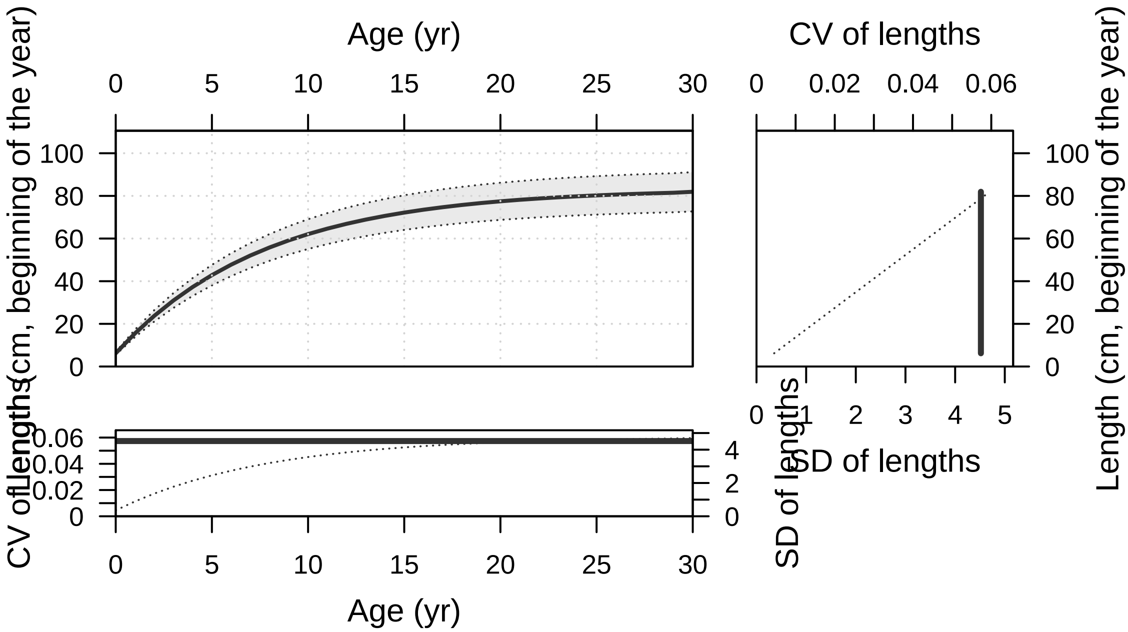
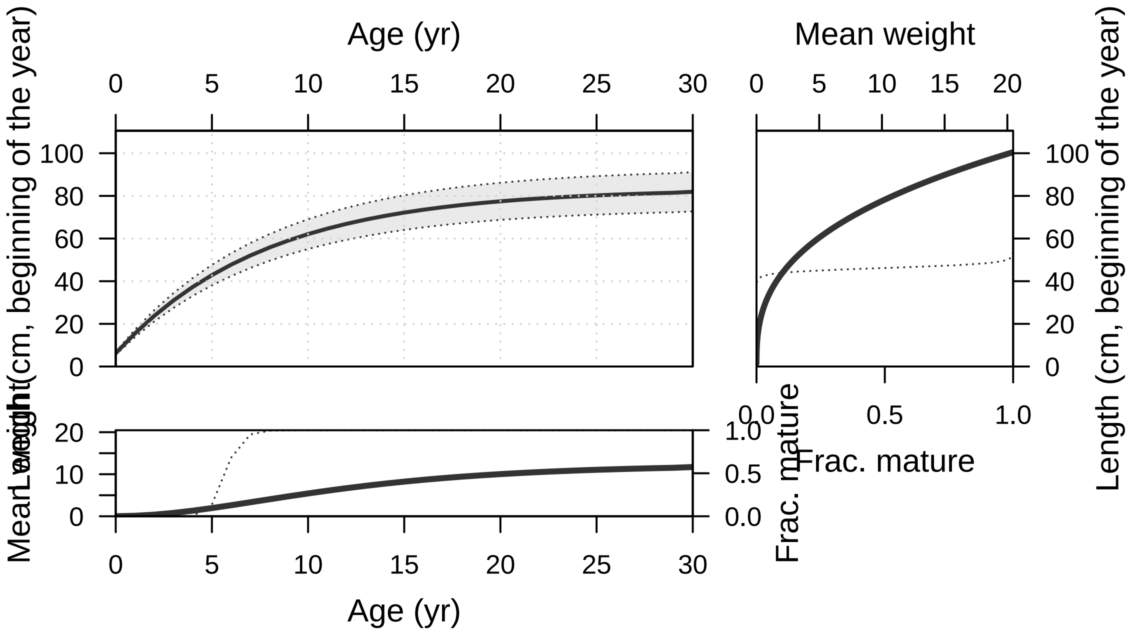
Plots created using the 'r4ss' package in R
Stock Synthesis version: 3.30.19.0
StartTime: Mon Feb 09 16:12:37 2026
Data_File: data.ss
Control_File: control.ss

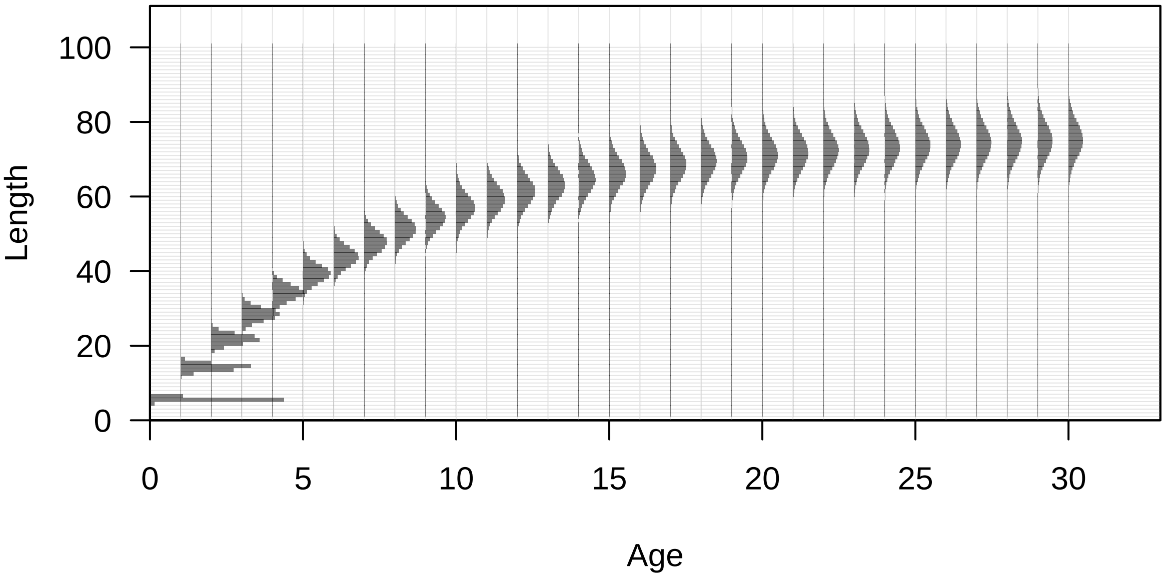

Length (cm, beginning of the year)
0
20
40
60
80
100
0
5
10
15
20
25
30
Age (yr)



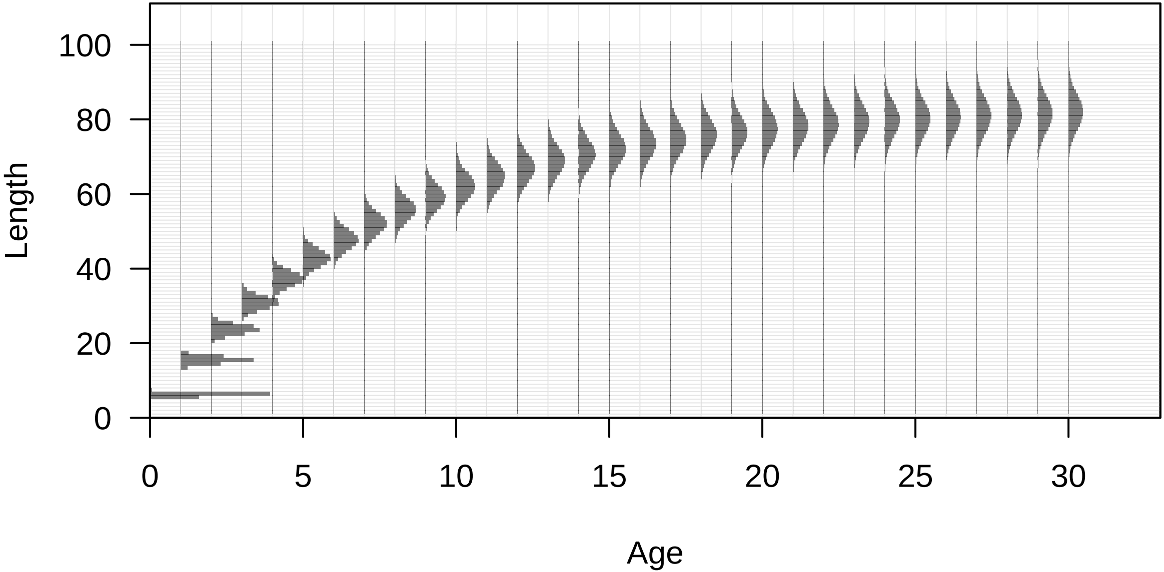
Age (yr)
CV of lengths
Length (cm, beginning of the year)
SD of lengths
CV of Lengths
SD of lengths
Age (yr)
0
5
10
15
20
25
30
0
0.02
0.04
0.06
100
80
60
40
20
0
1
2
3
4
5
0.06
0.04
0.02
4
2



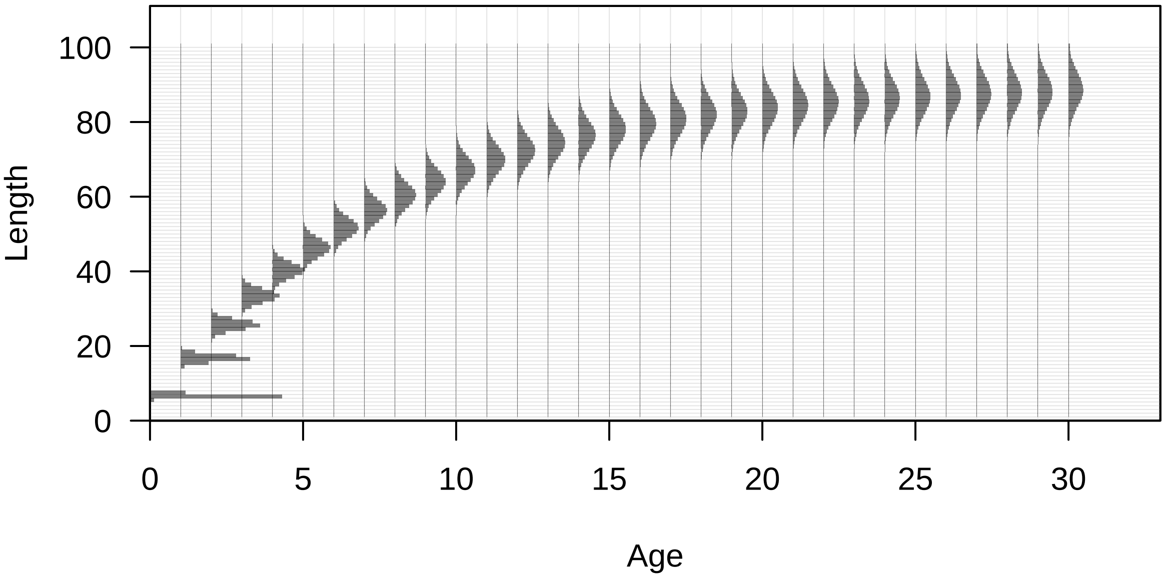
Age (yr)
Mean weight
Length (cm, beginning of the year)
Mean weight
Frac. mature
Frac. mature
Age (yr)
0
5
10
15
20
25
30
100
80
60
40
20
0
0.0
0.5
1.0

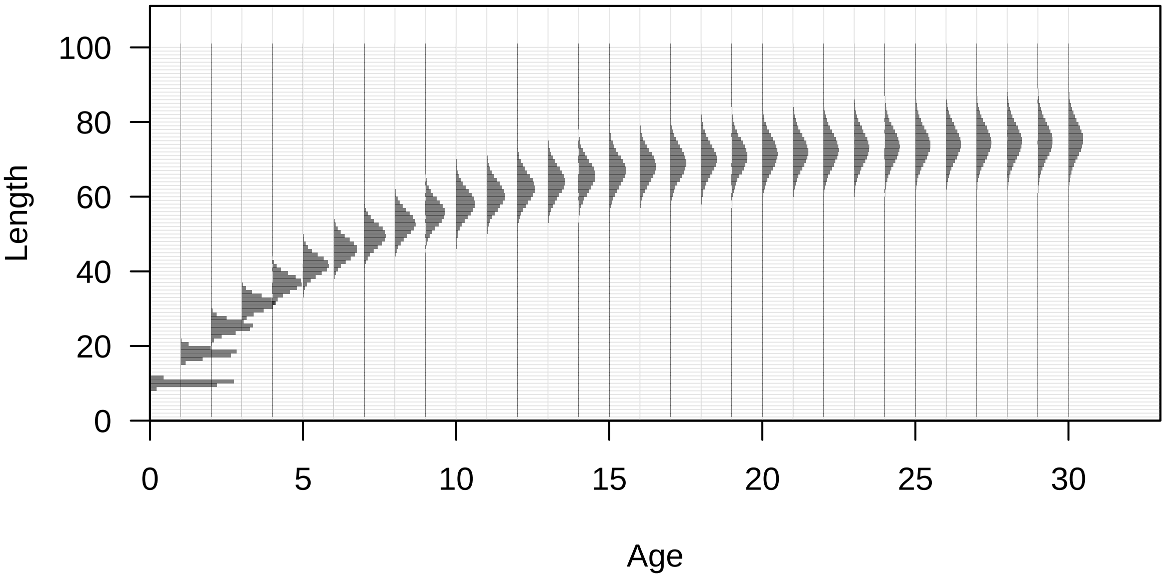


Length
0
20
40
60
80
100
0
5
10
15
20
25
30
Age

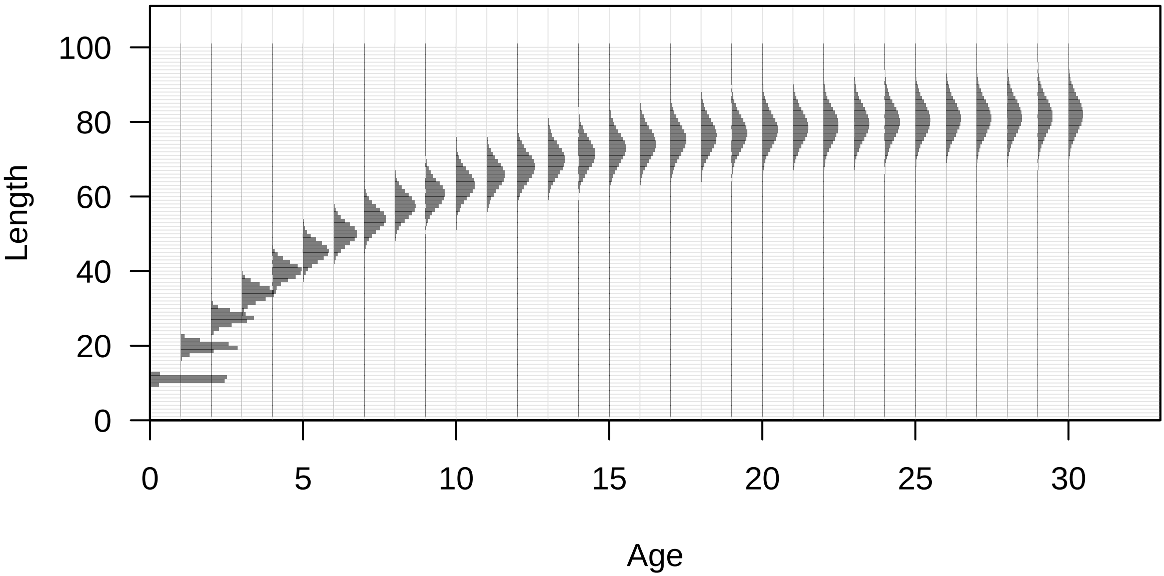


Length
Age
0
20
40
60
80
100
0
5
10
15
20
25
30

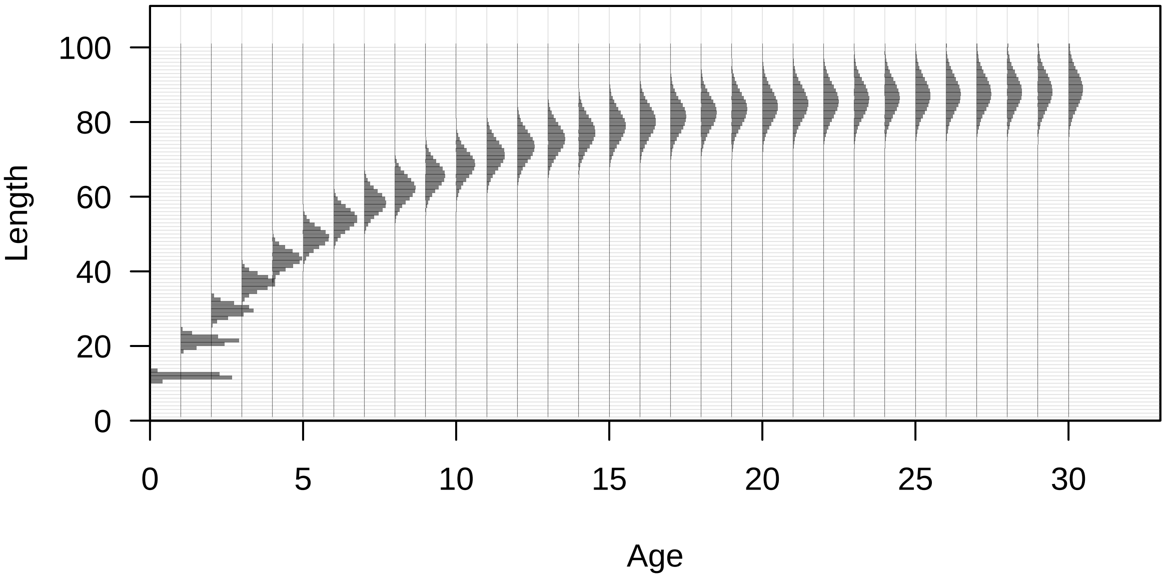

Length
0
20
40
60
80
100
Age
0
5
10
15
20
25
30



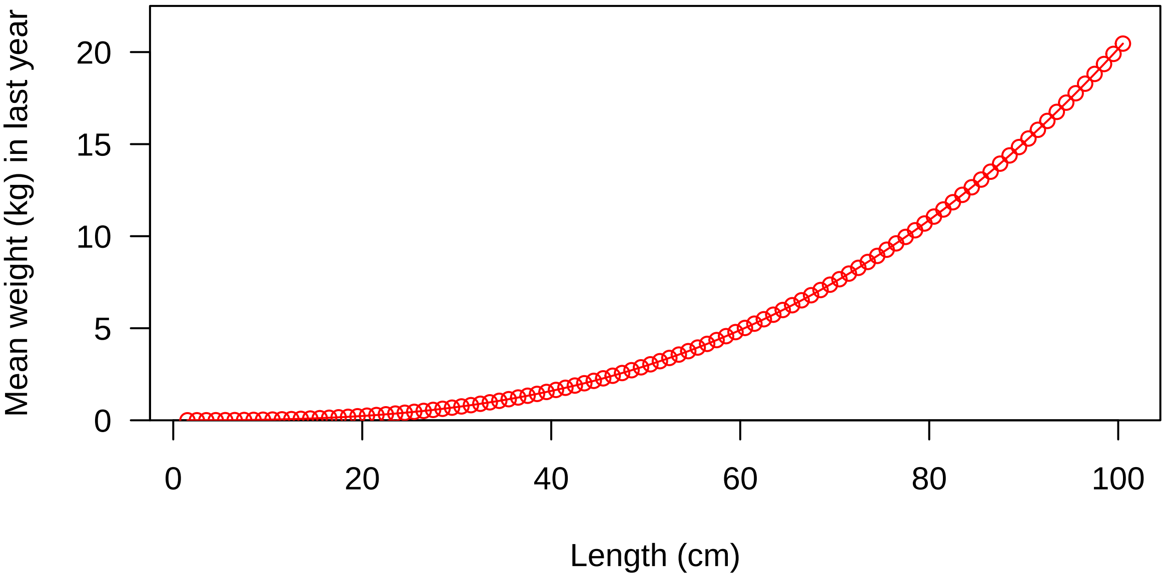
Length
100
80
60
40
20
0
0
5
10
15
20
25
30
Age

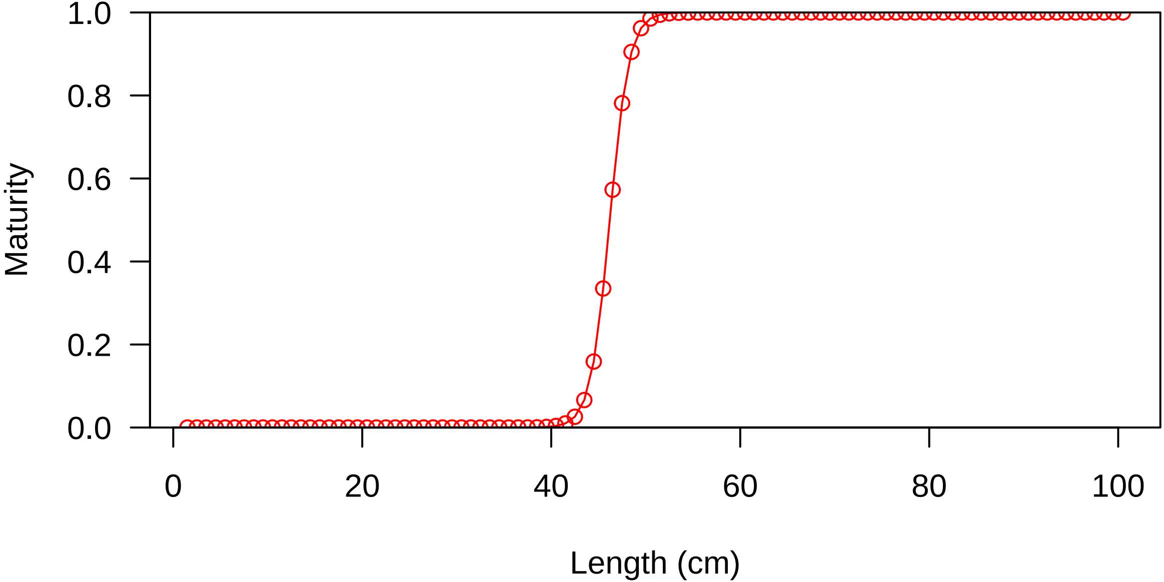


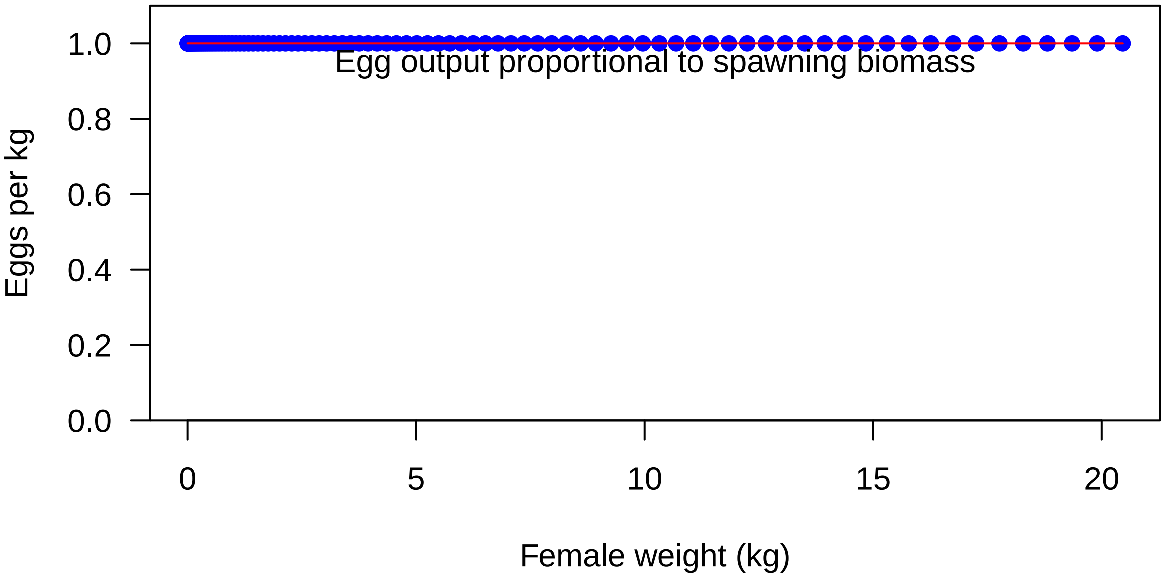
Length
Age
0
20
40
60
80
100
0
5
10
15
20
25
30



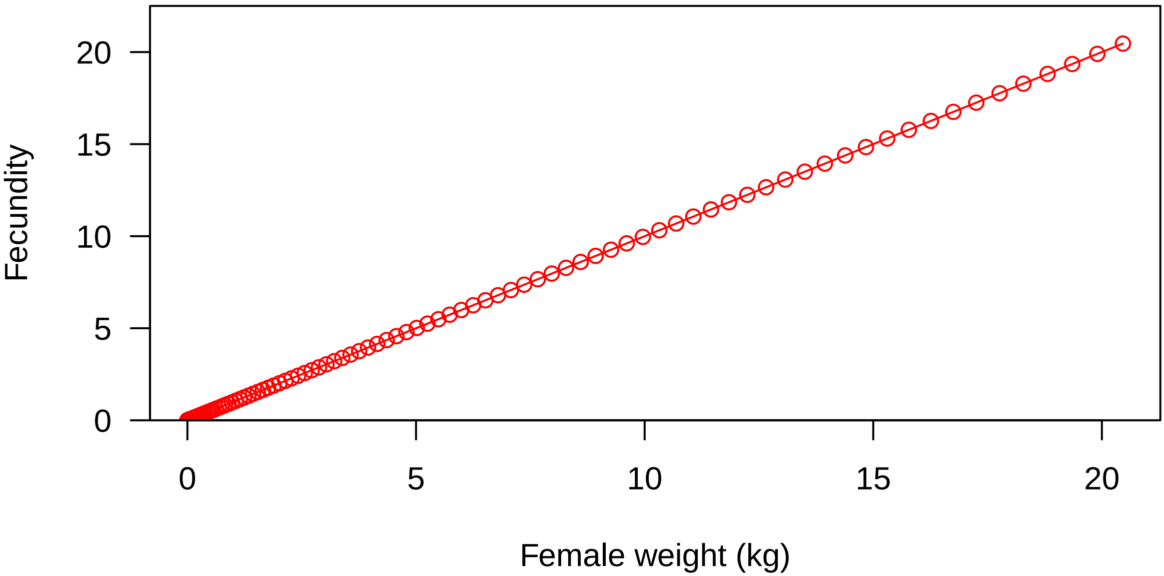
Length
100
80
60
40
20
0
0
5
10
15
20
25
30
Age

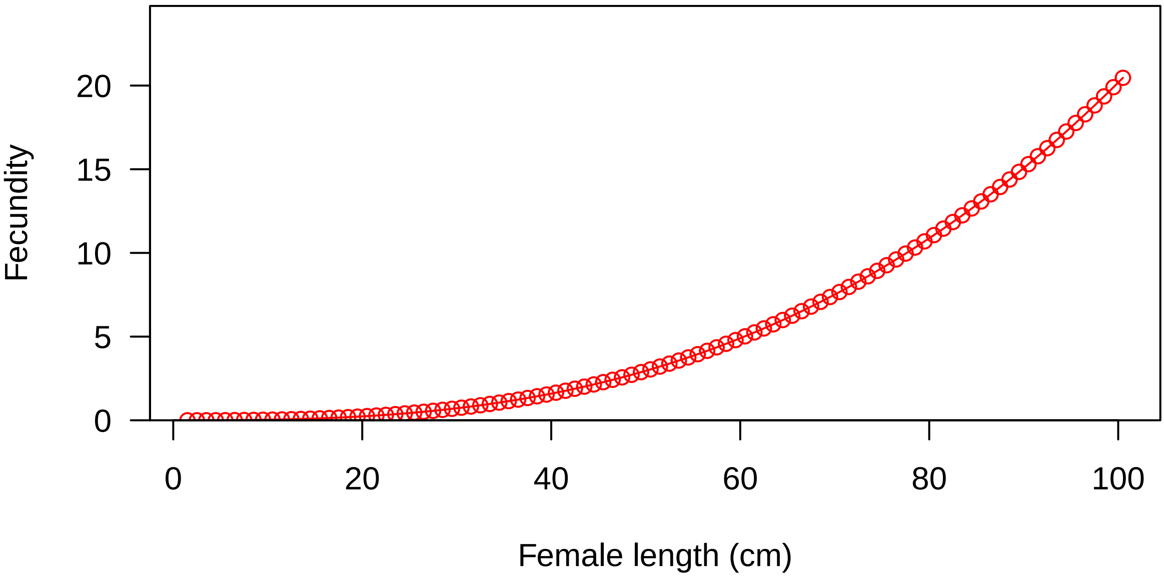

Mean weight (kg) in last year
0
5
10
15
20
0
20
40
60
80
100
Length (cm)

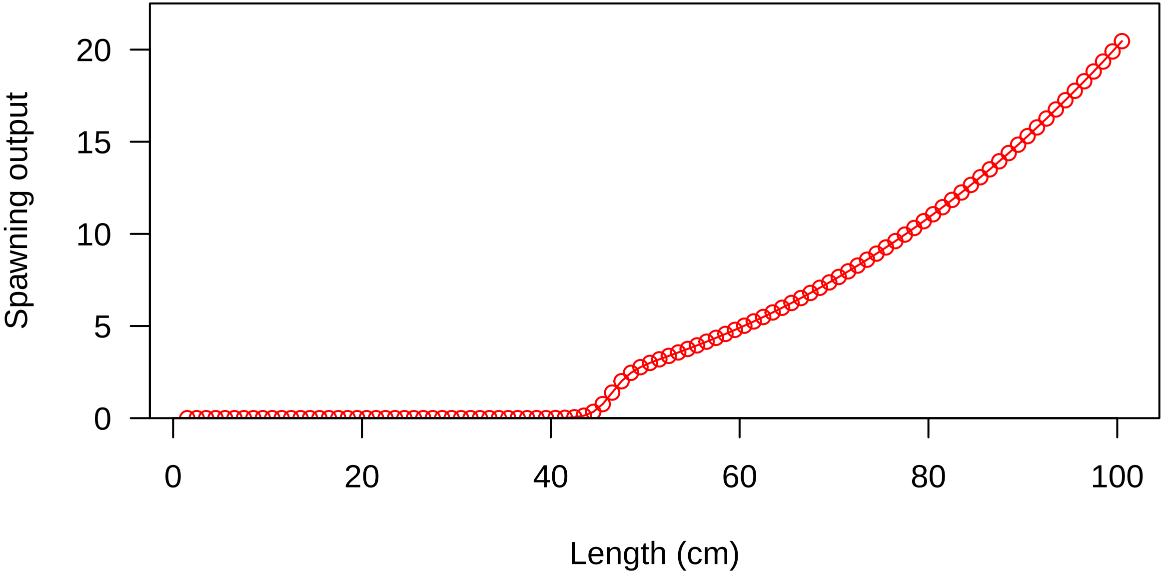

1.0
0.8
0.6
0.4
0.2
0.0
Maturity
0
20
40
60
80
100
Length (cm)

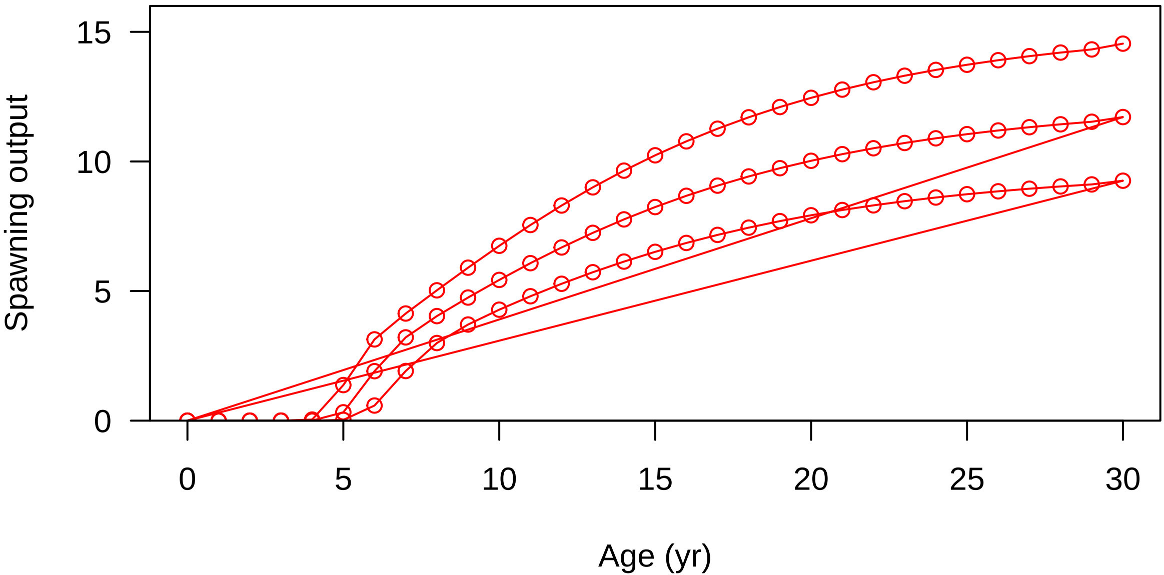


Egg output proportional to spawning biomass
1.0
0.8
0.6
0.4
0.2
0.0
Eggs per kg
0
5
10
15
20
Female weight (kg)

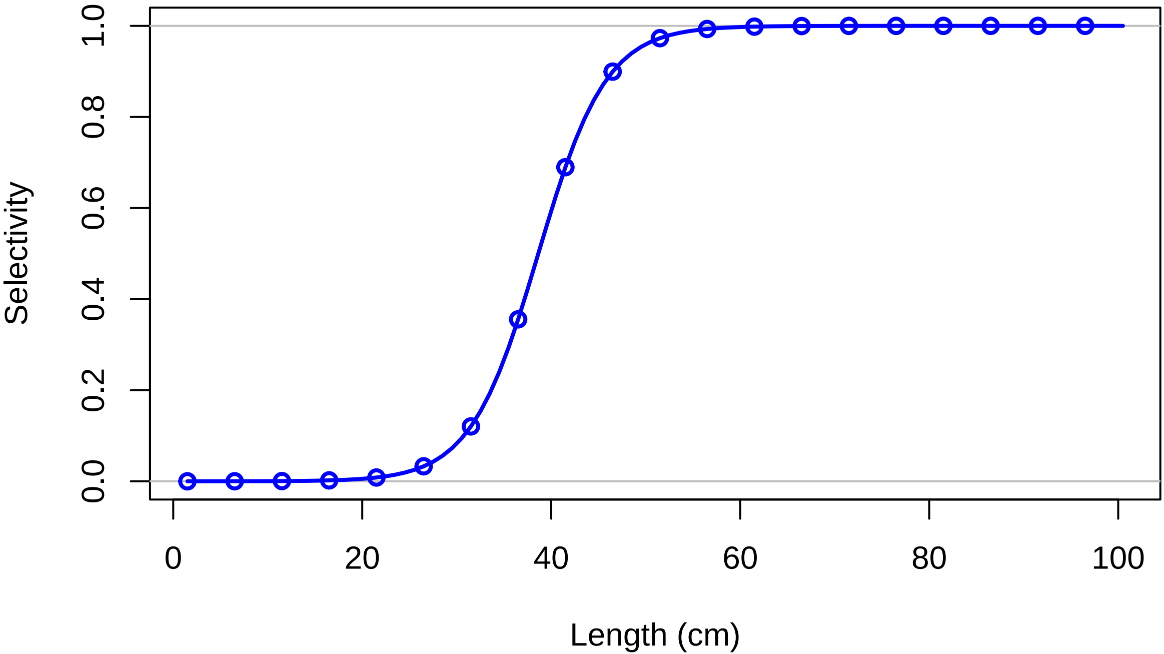

Fecundity
0
5
10
15
20
0
5
10
15
20
Female weight (kg)

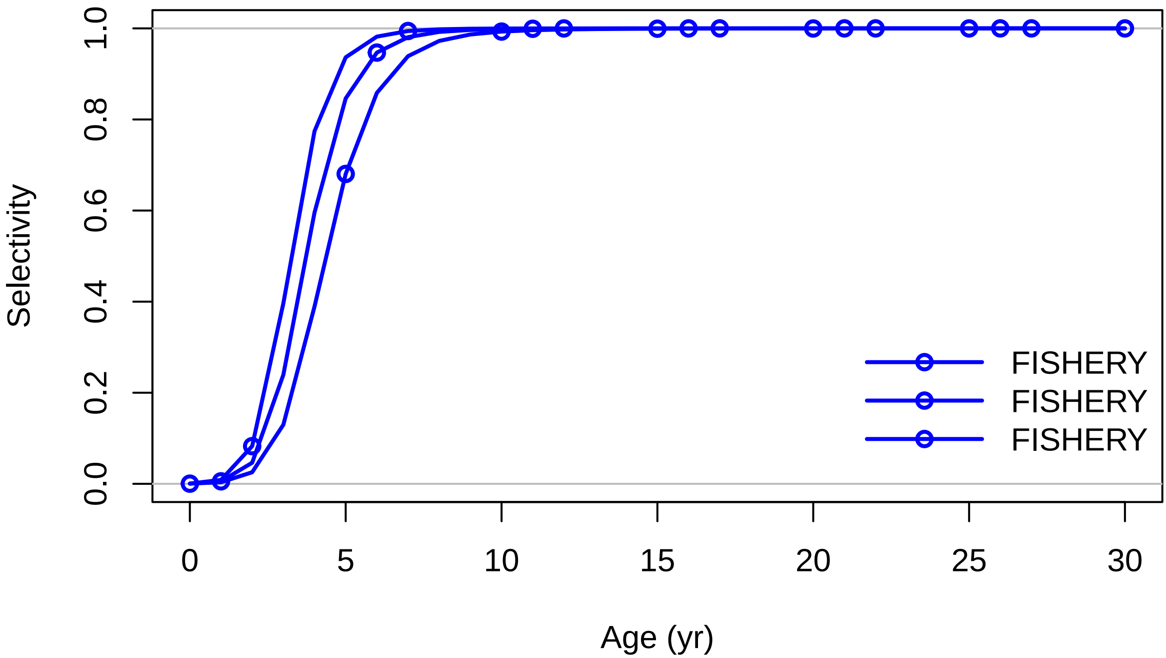

Fecundity
0
5
10
15
20
0
20
40
60
80
100
Female length (cm)


Spawning output
0
5
10
15
20
0
20
40
60
80
100
Length (cm)

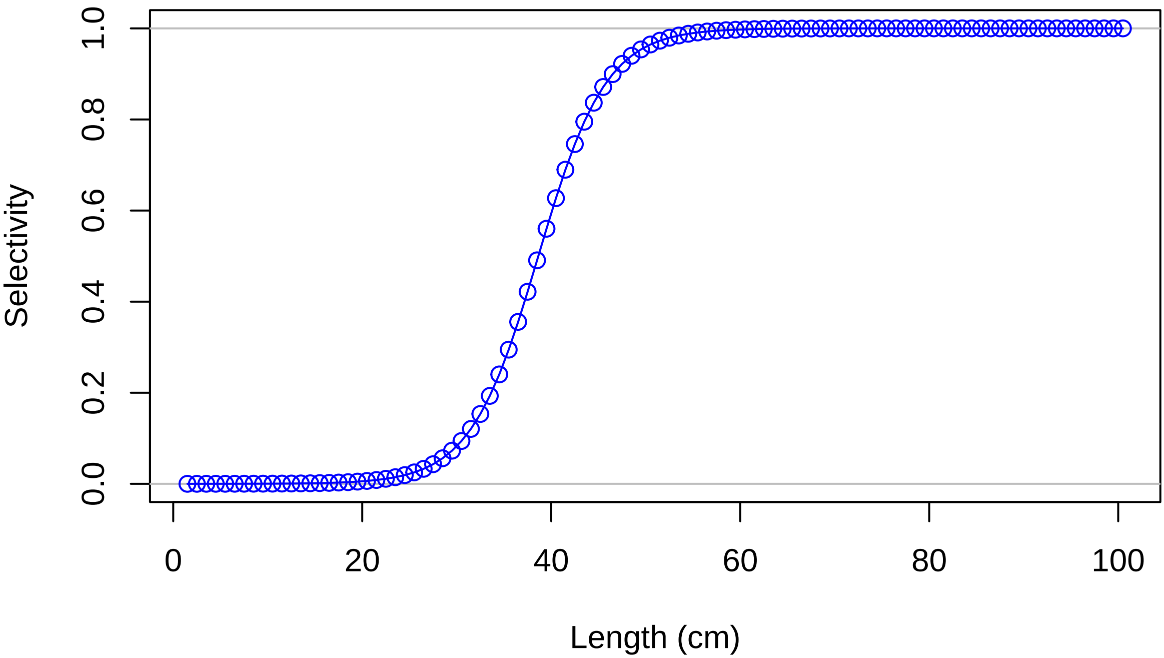


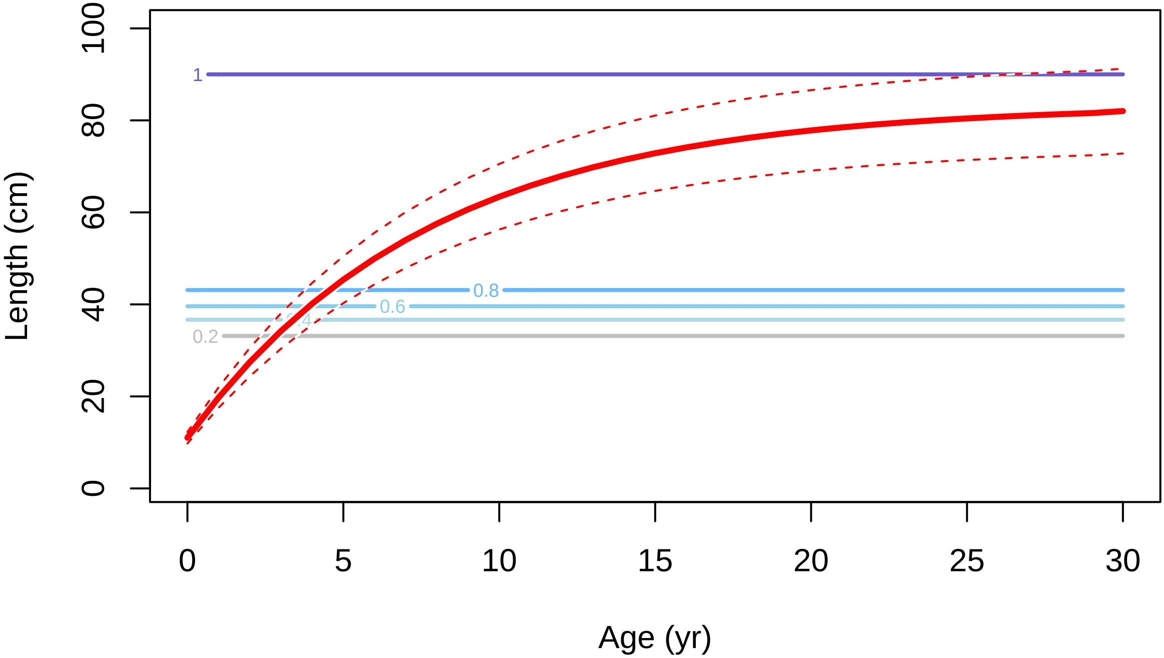
Spawning output
15
10
5
0
0
5
10
15
20
25
30
Age (yr)


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
Length (cm)

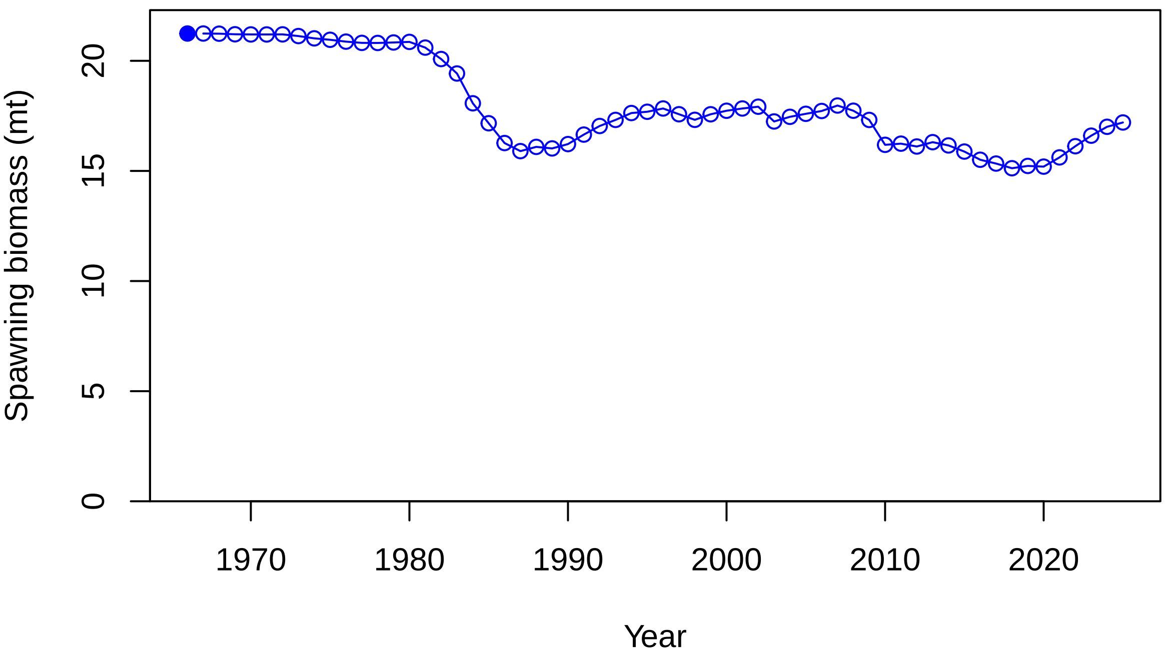


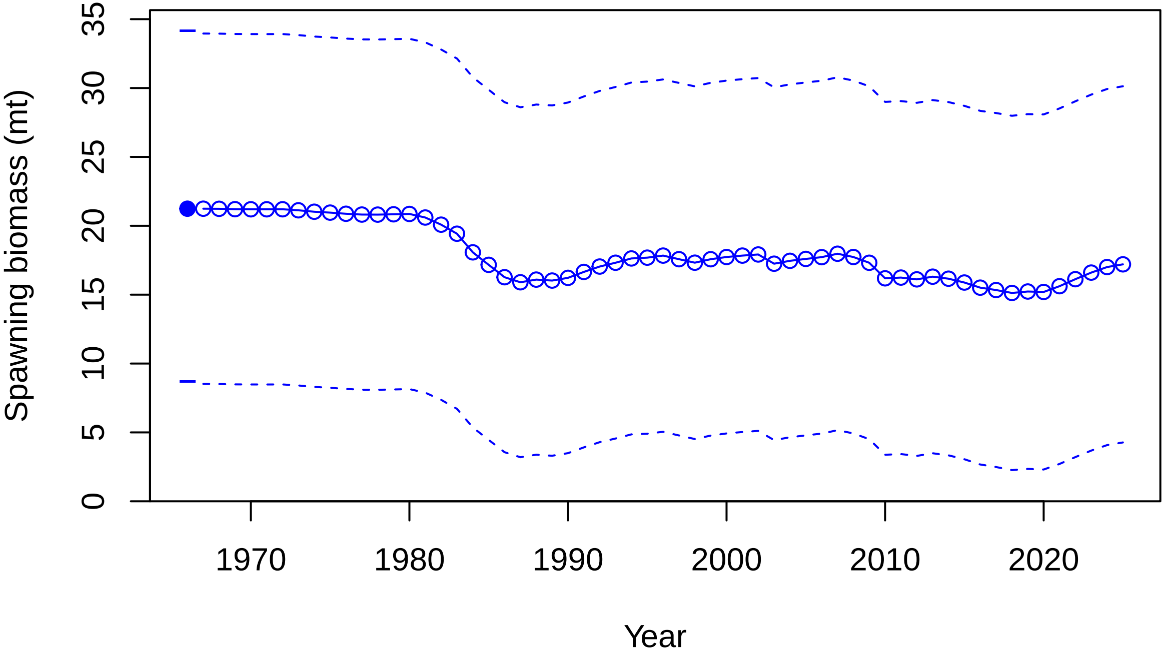
Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30
Age (yr)
FISHERY
FISHERY
FISHERY


Selectivity
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Length (cm)

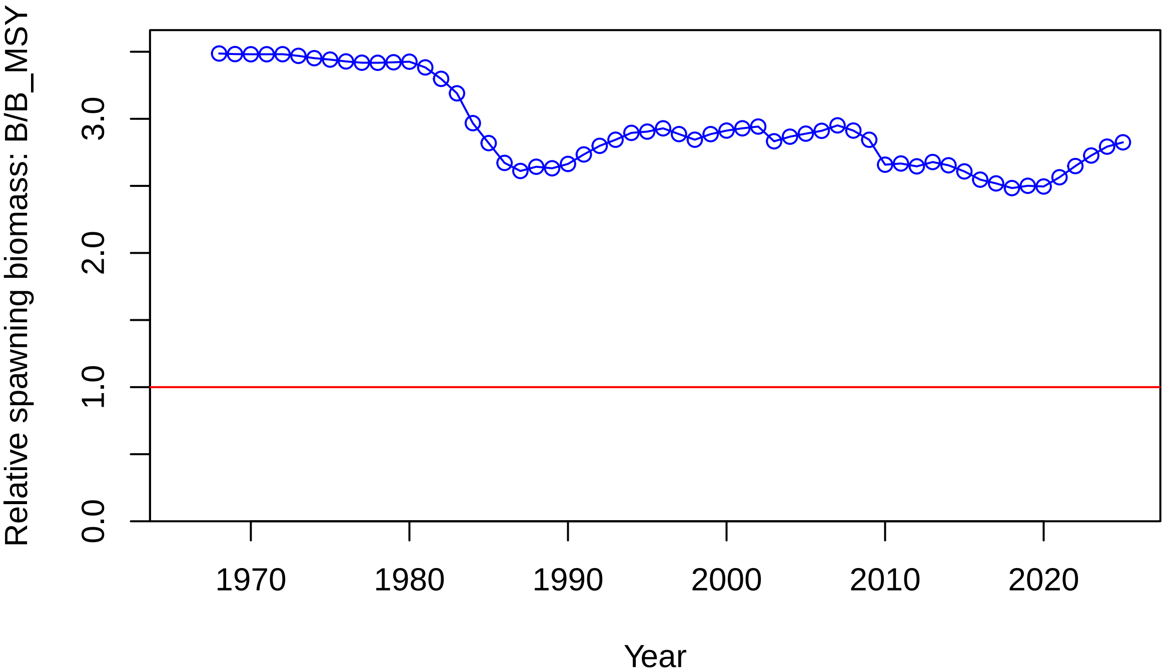


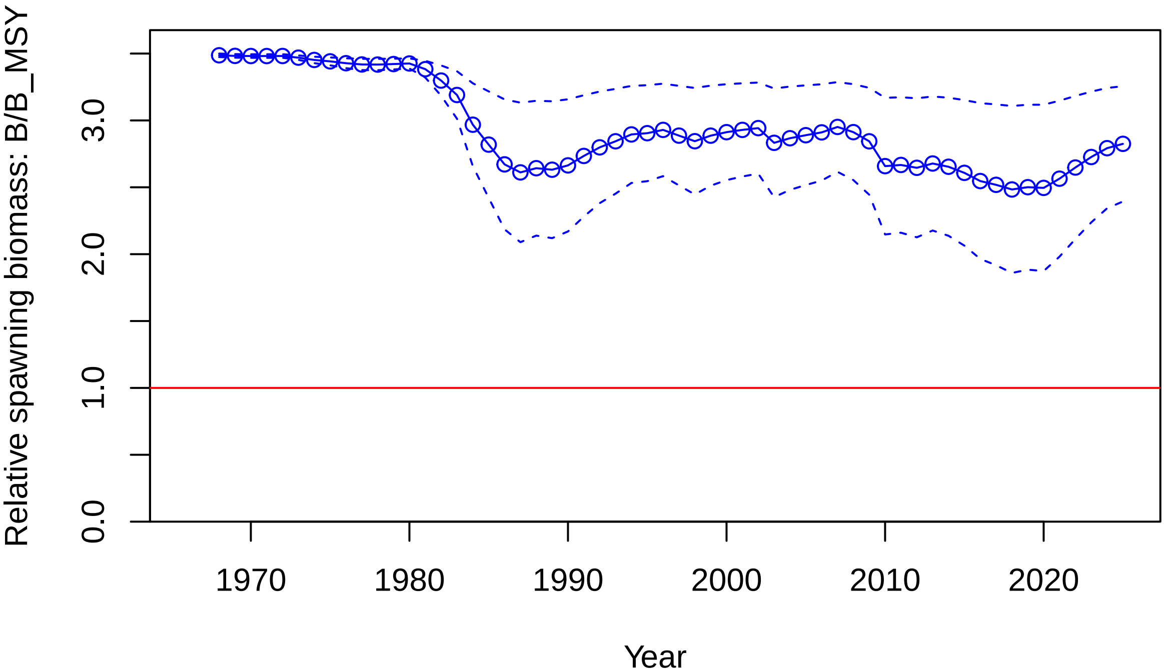
Length (cm)
Age (yr)
0
20
40
60
80
100
0
5
10
15
20
25
30
1
0.8
0.6
0.4
0.2

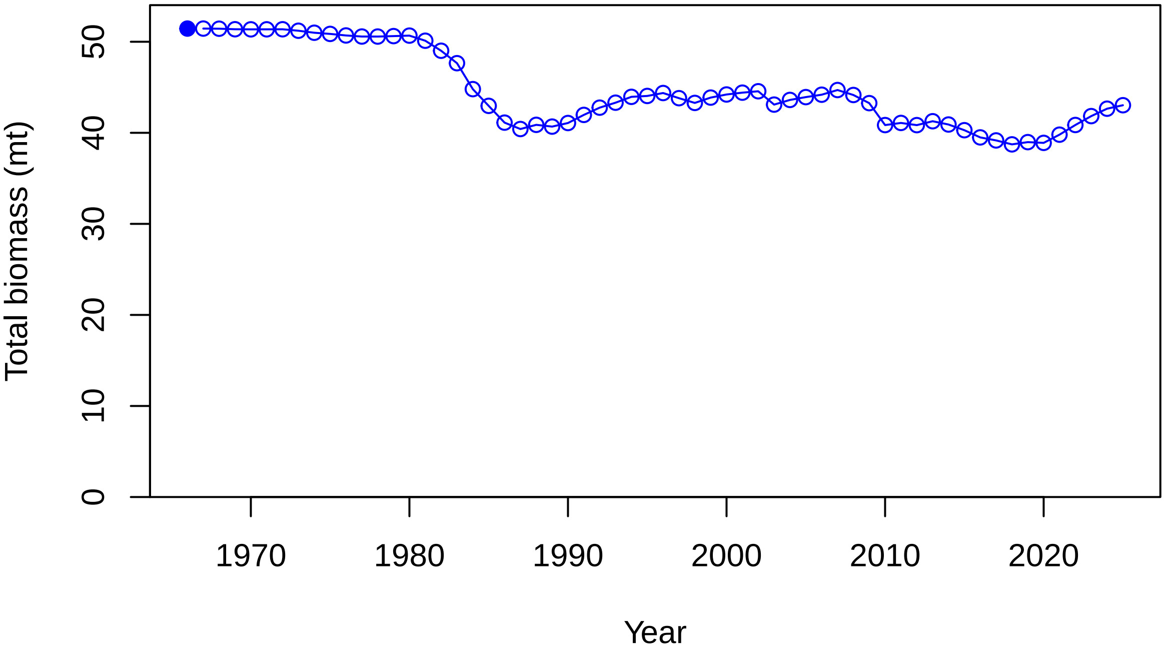

Spawning biomass (mt)
0
5
10
15
20
1970
1980
1990
2000
2010
2020
Year



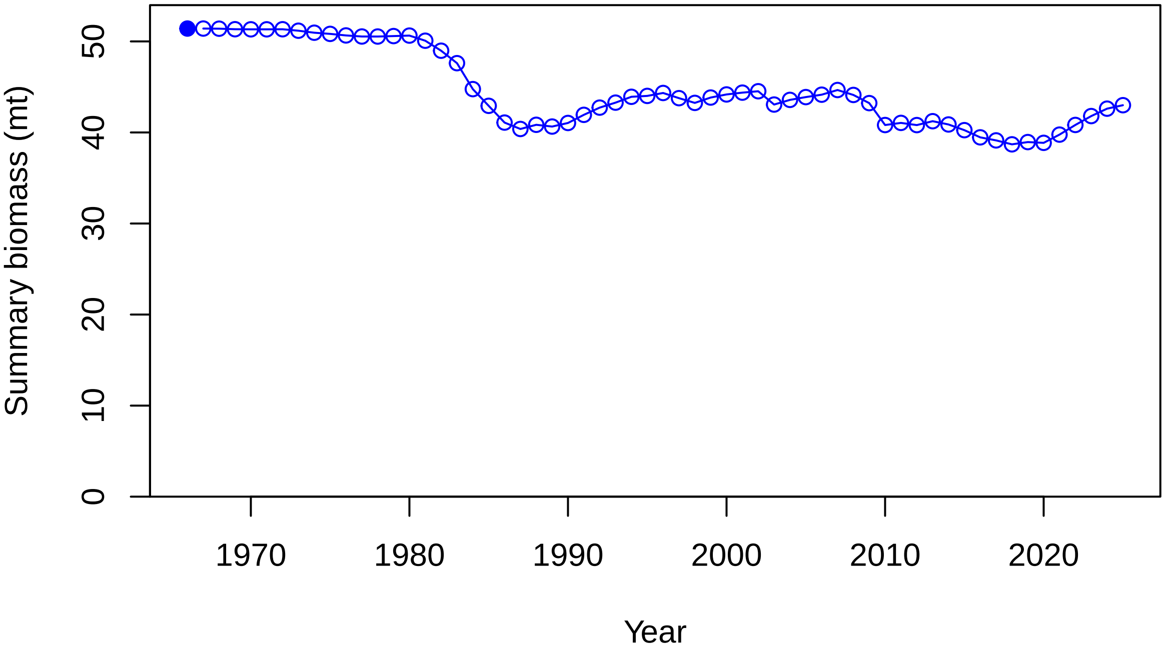
Spawning biomass (mt)
0
5
10
15
20
25
30
35
1970
1980
1990
2000
2010
2020
Year


Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year

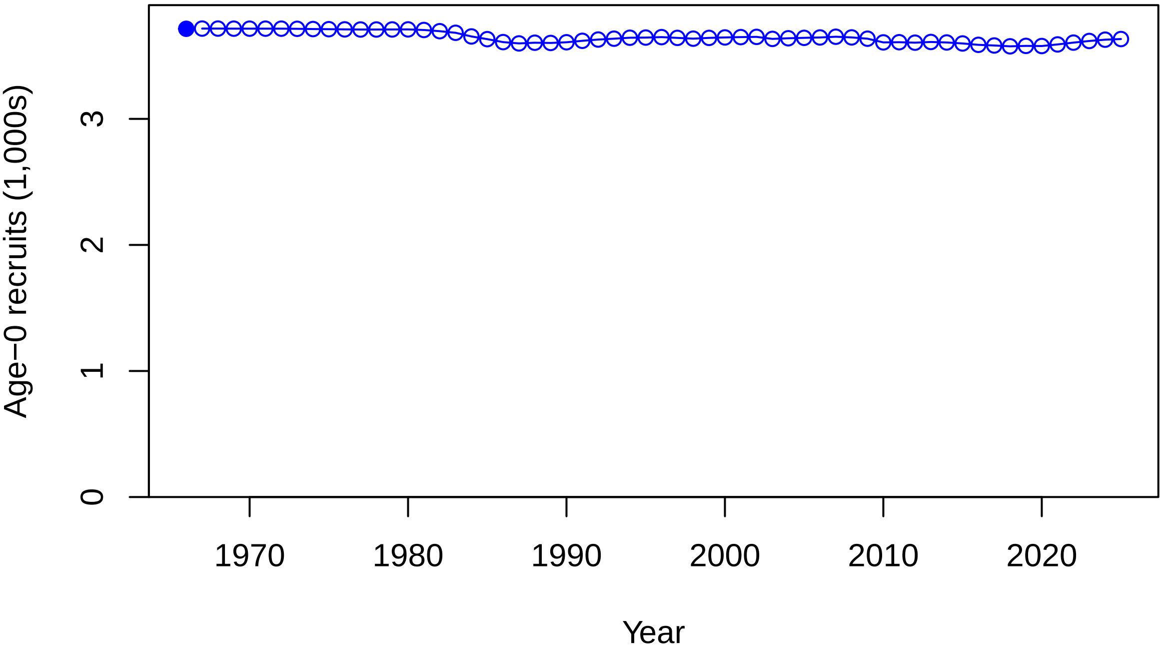

Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year



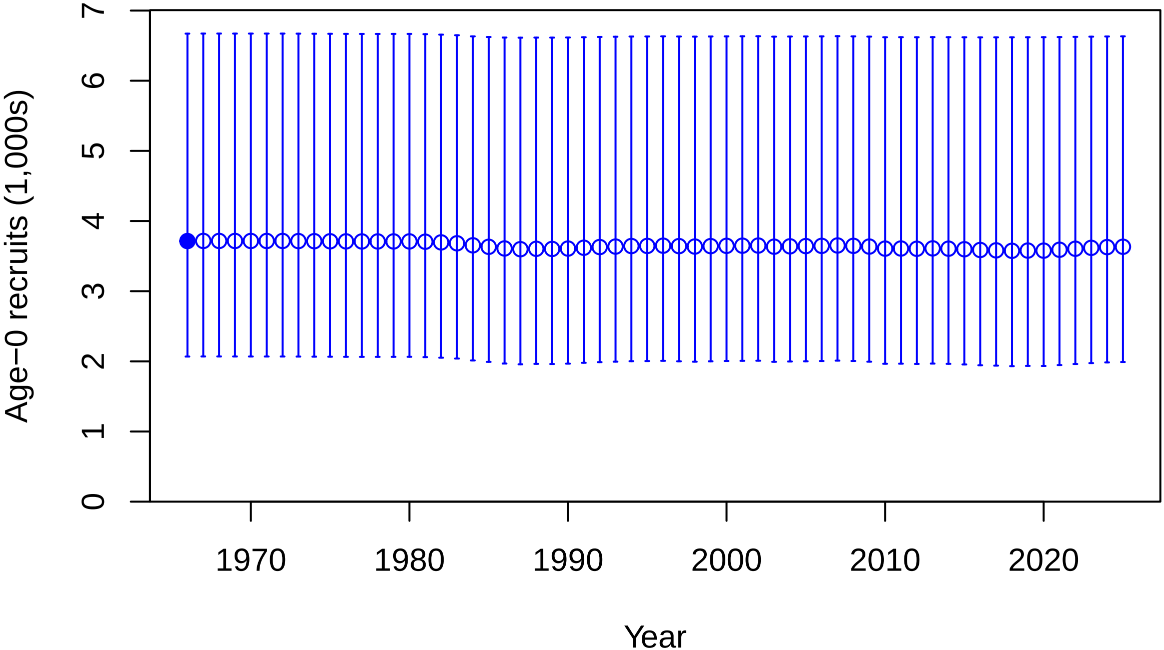
Total biomass (mt)
0
10
20
30
40
50
1970
1980
1990
2000
2010
2020
Year

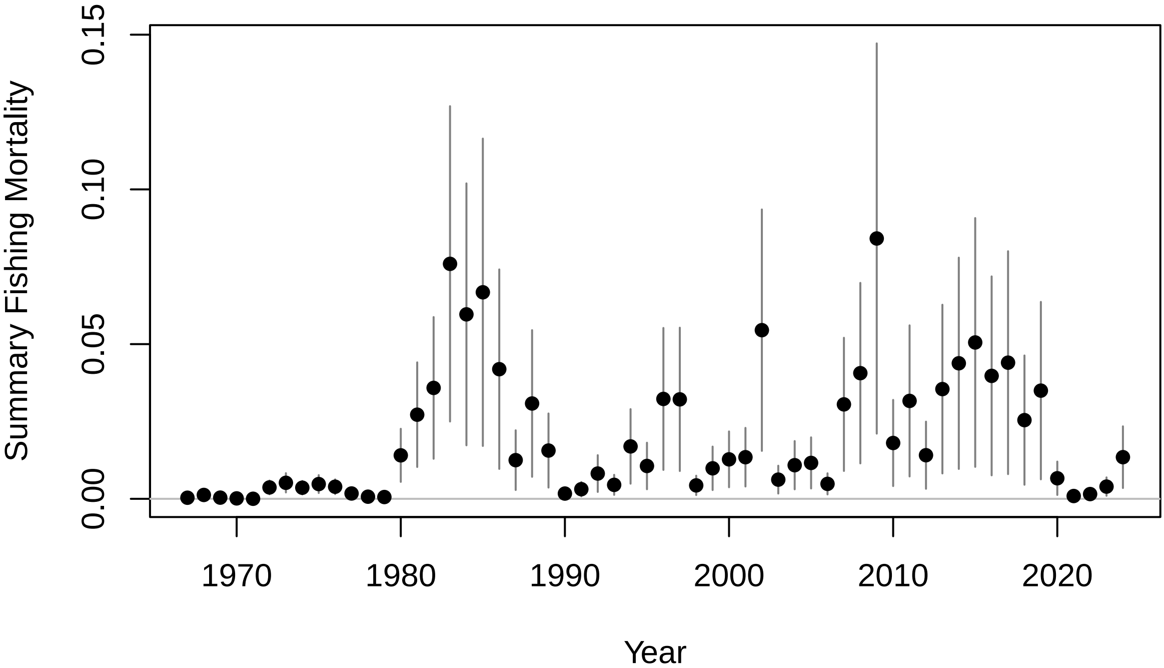


Summary biomass (mt)
0
10
20
30
40
50
1970
1980
1990
2000
2010
2020
Year

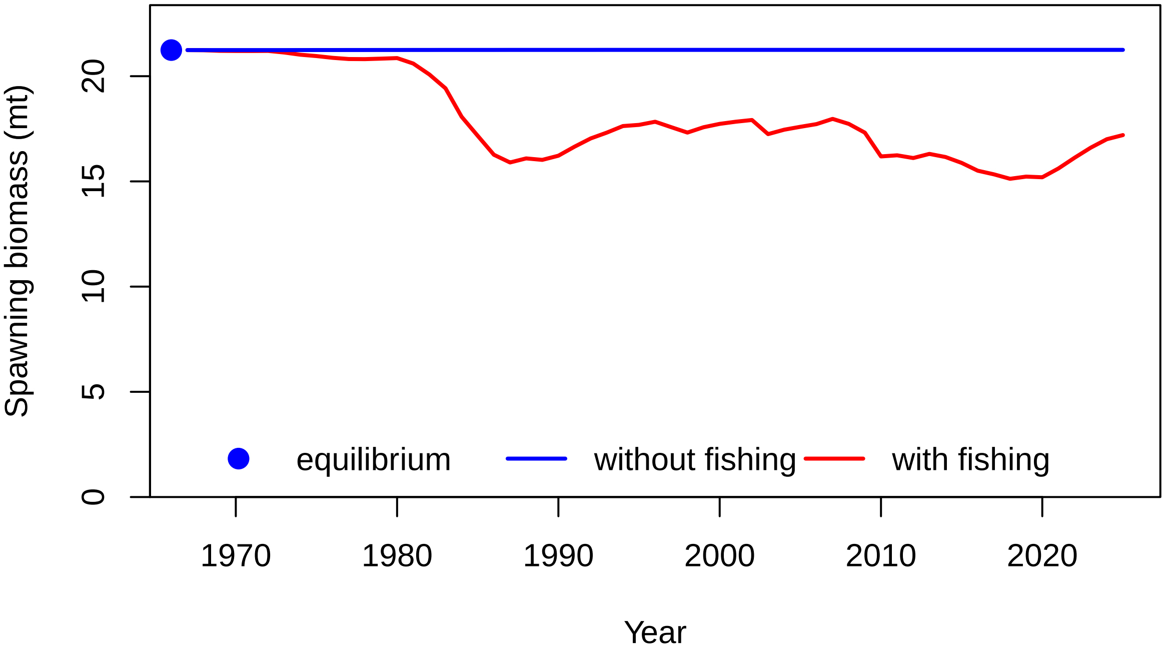


Age−0 recruits (1,000s)
0
1
2
3
1970
1980
1990
2000
2010
2020
Year

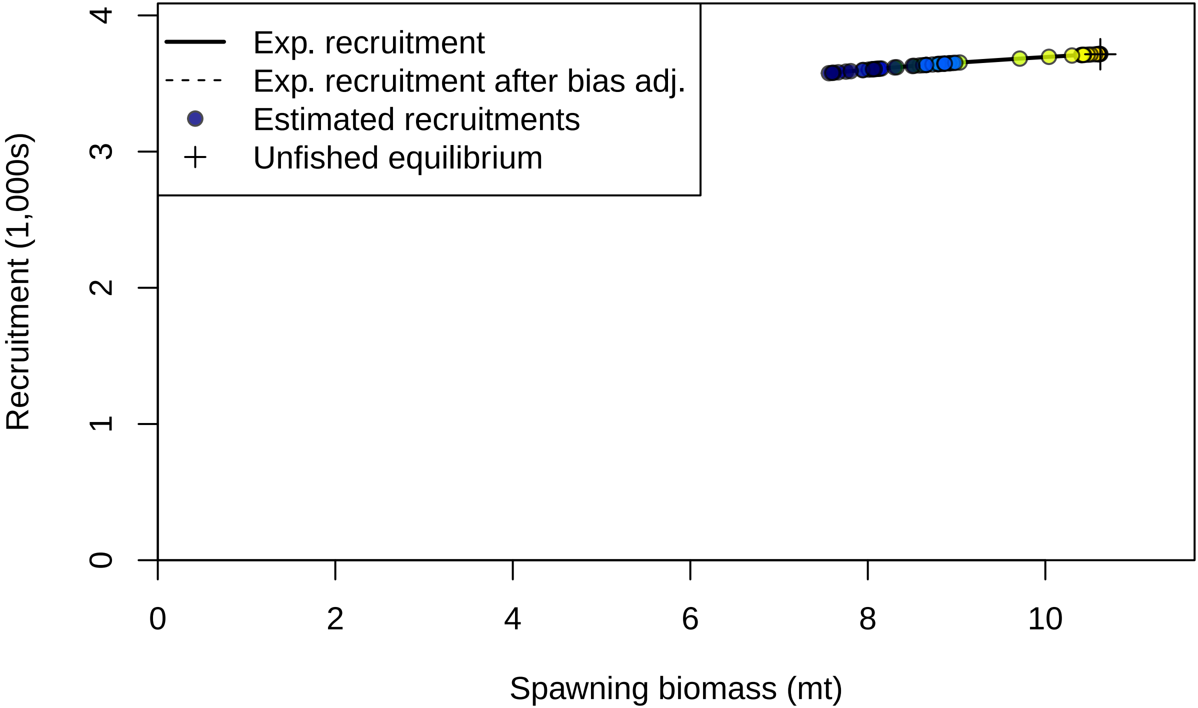

Age−0 recruits (1,000s)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year



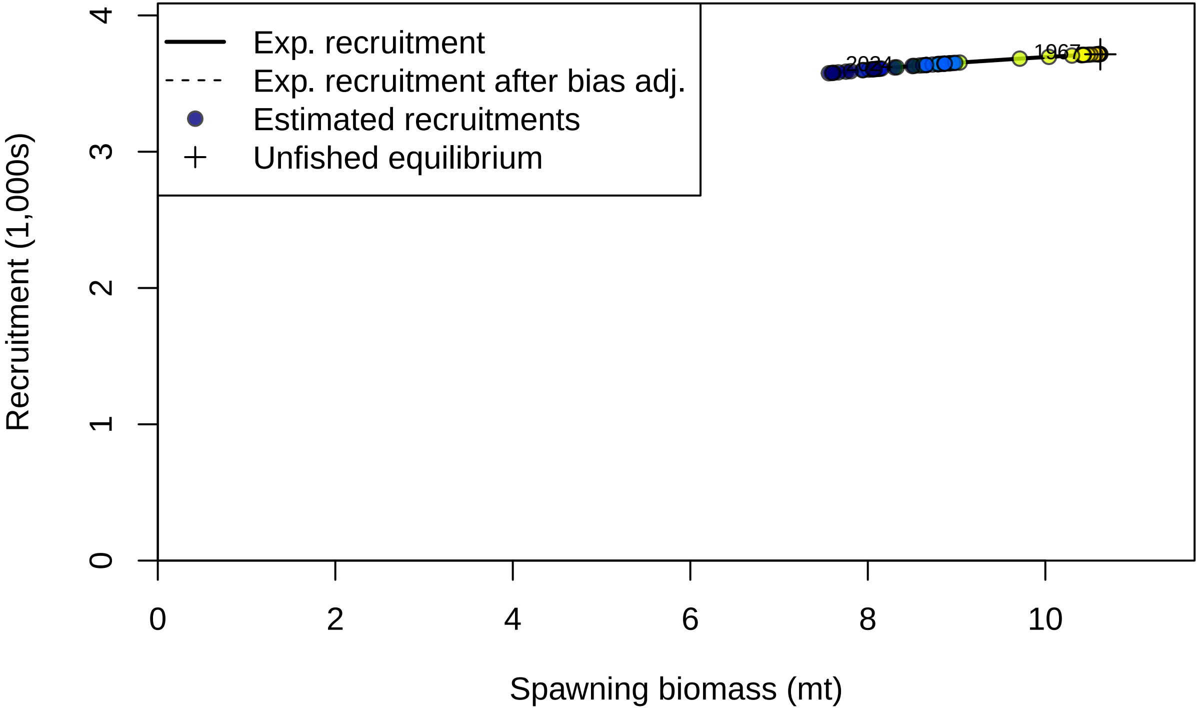
Summary Fishing Mortality
0.00
0.05
0.10
0.15
1970
1980
1990
2000
2010
2020
Year

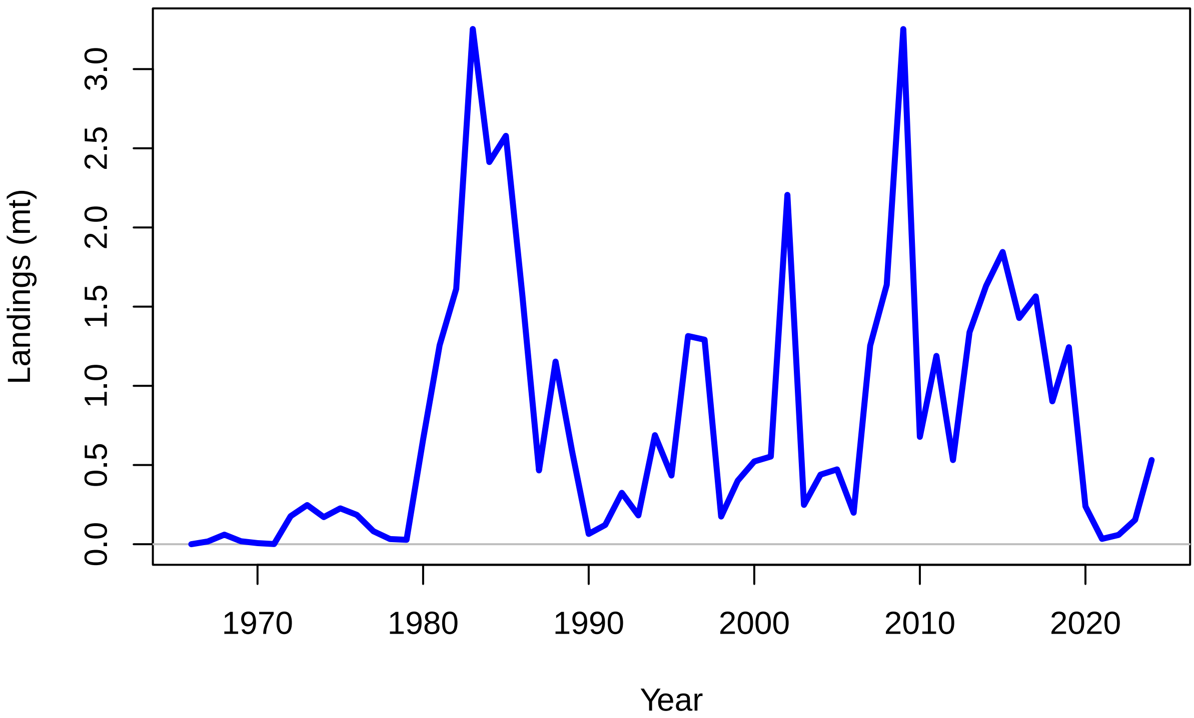


Spawning biomass (mt)
0
5
10
15
20
1970
1980
1990
2000
2010
2020
Year
equilibrium
without fishing
with fishing

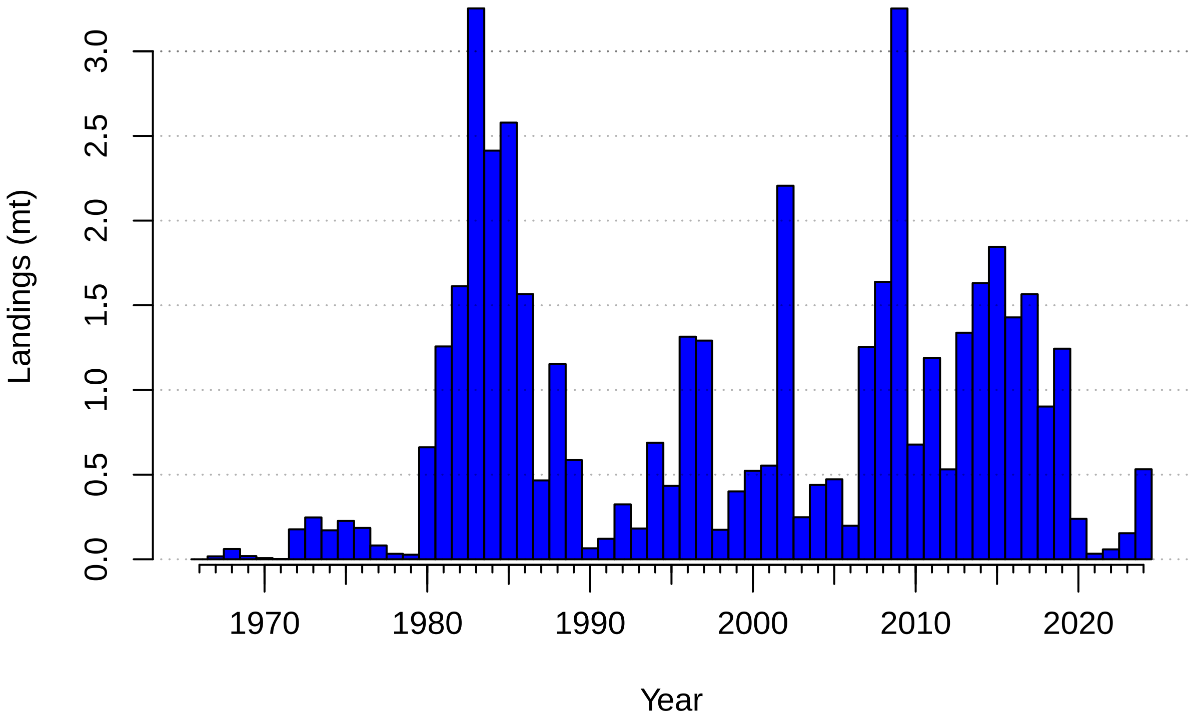


Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium
Recruitment (1,000s)
Spawning biomass (mt)
0
1
2
3
4
0
2
4
6
8
10

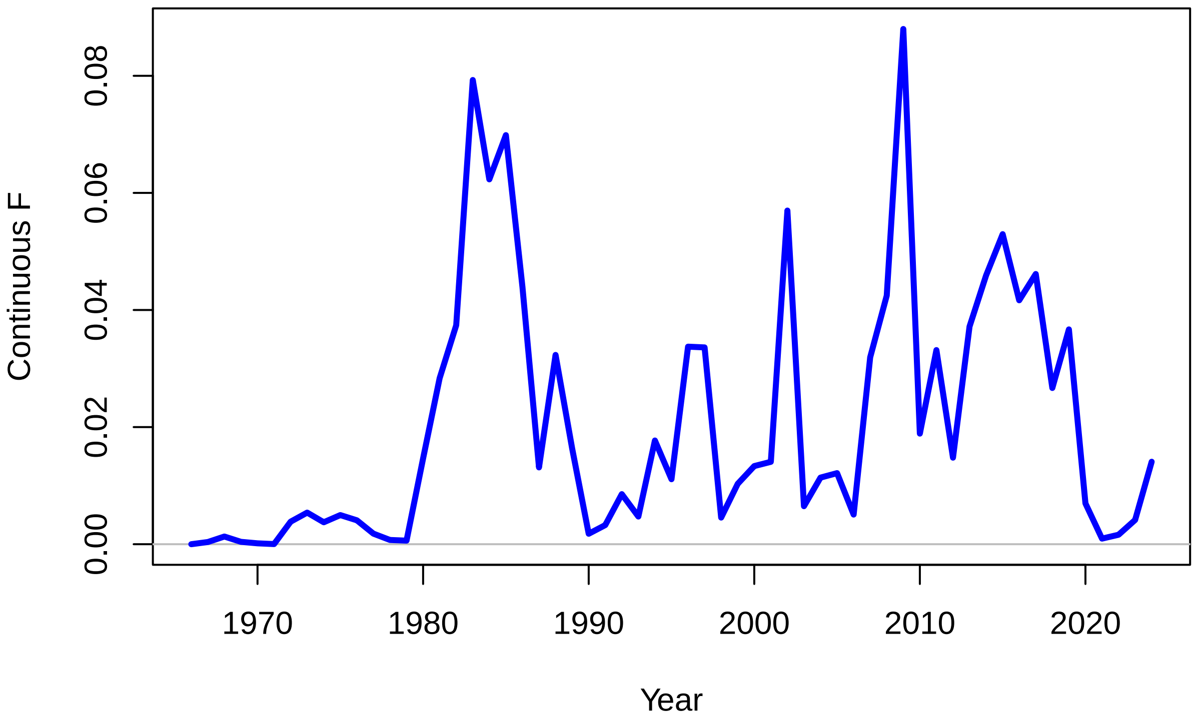

Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium
2024
1967
Recruitment (1,000s)
Spawning biomass (mt)
0
1
2
3
4
0
2
4
6
8
10



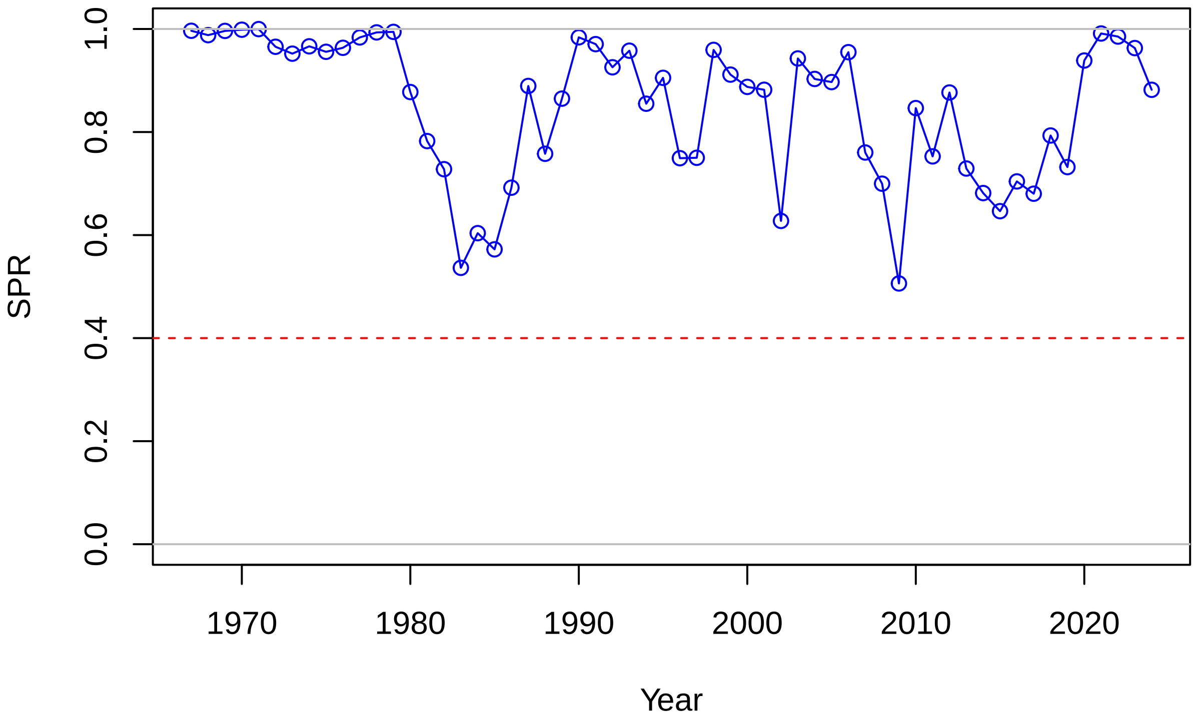
Landings (mt)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year

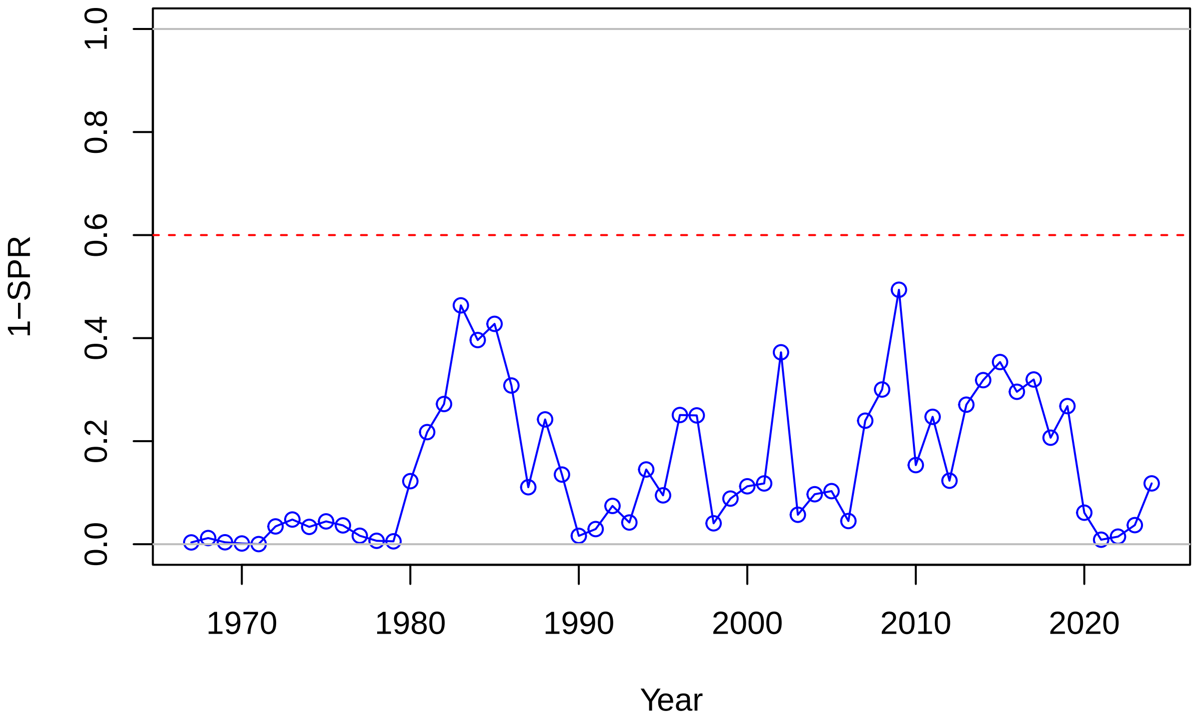


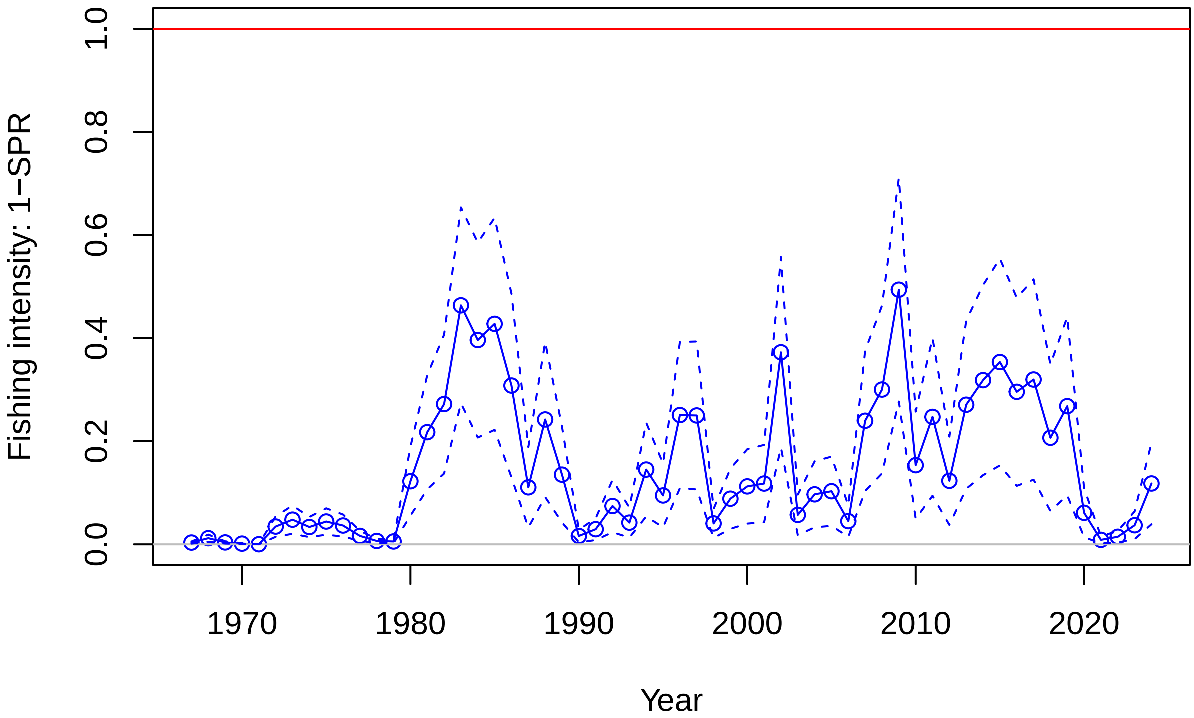
Landings (mt)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year



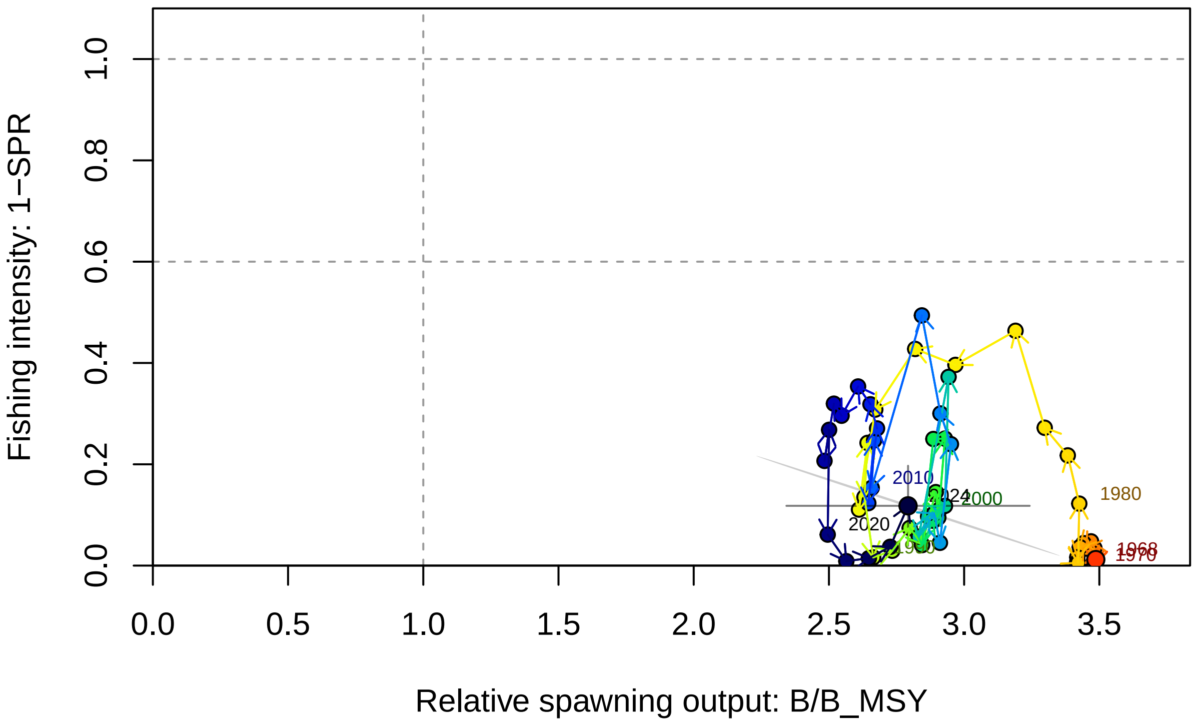
Continuous F
0.00
0.02
0.04
0.06
0.08
1970
1980
1990
2000
2010
2020
Year

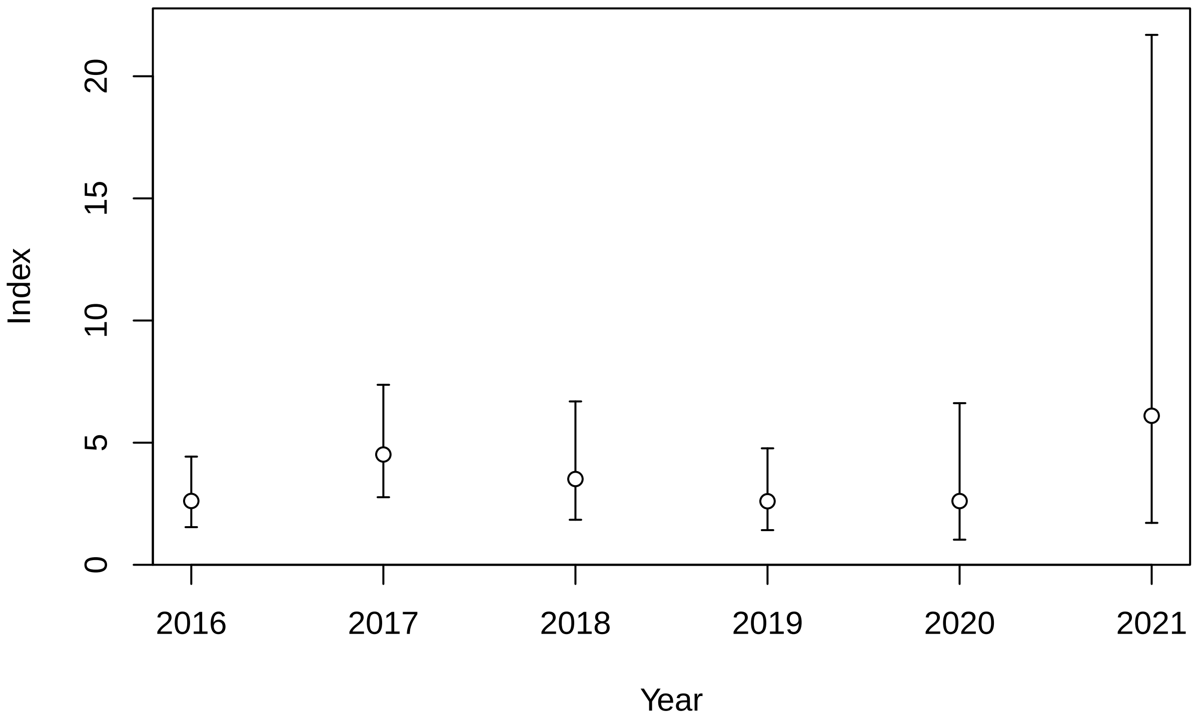

SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

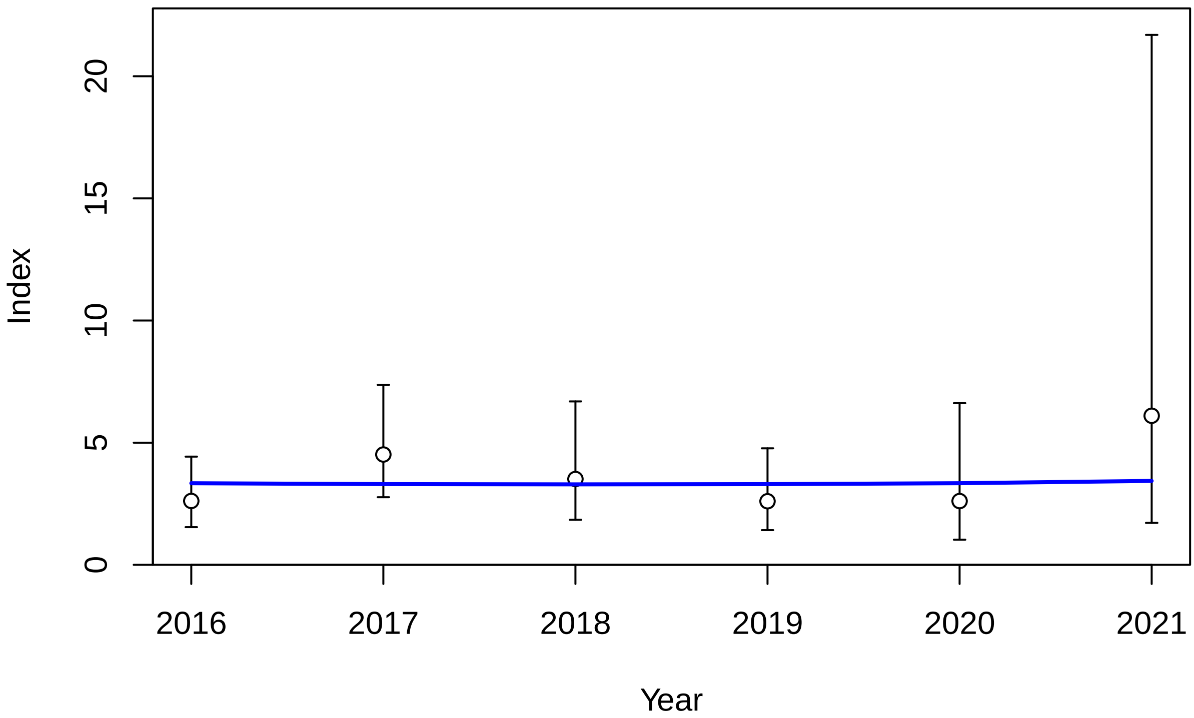

1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year



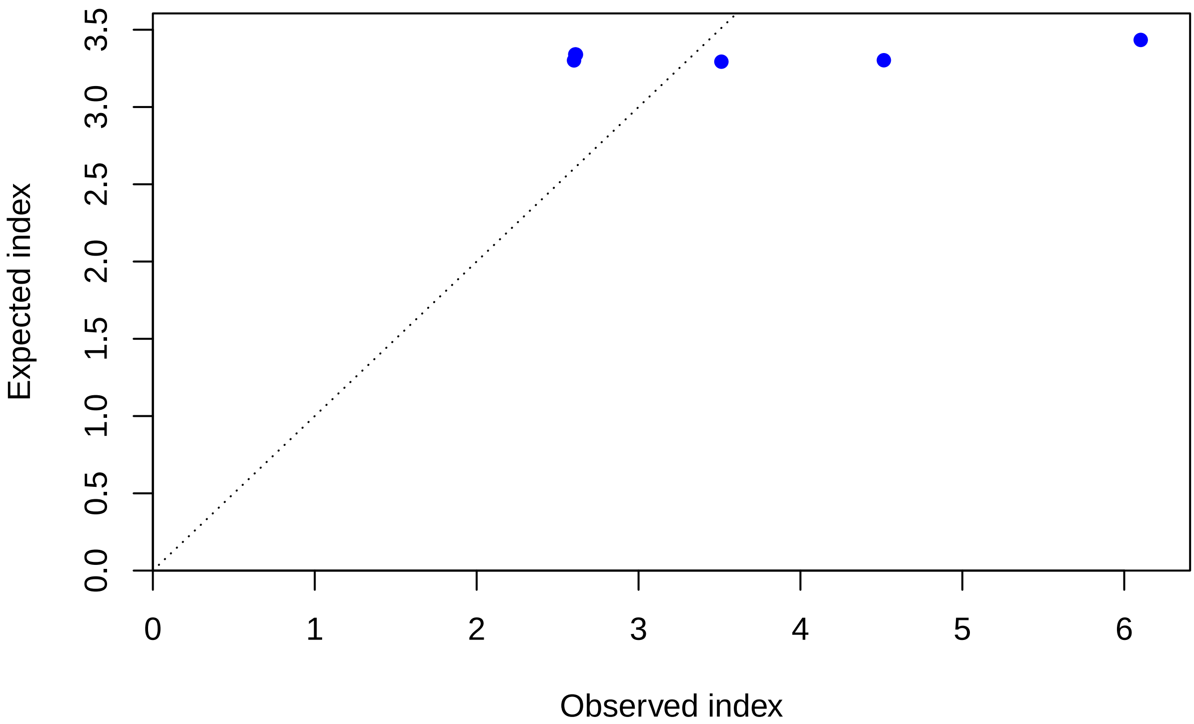
Fishing intensity: 1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year


Fishing intensity: 1−SPR
Relative spawning output: B/B_MSY
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
2010
2024
2000
2020
1980
1968
1970

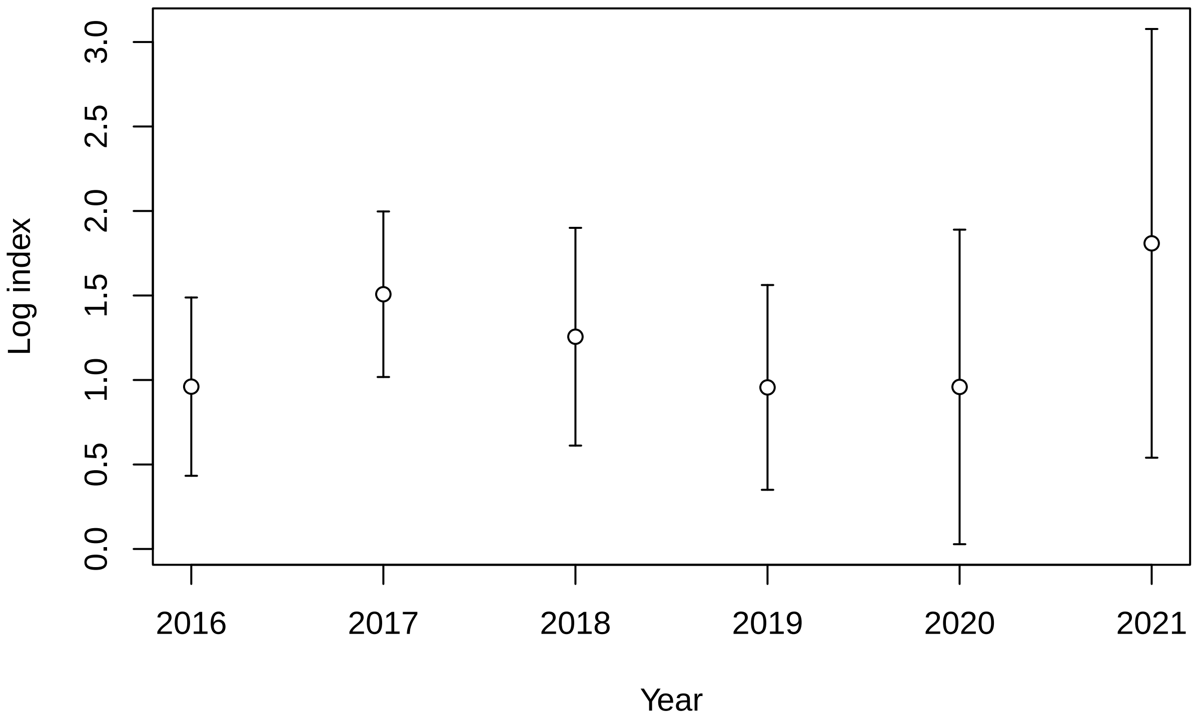

Index
Year
0
5
10
15
20
2016
2017
2018
2019
2020
2021



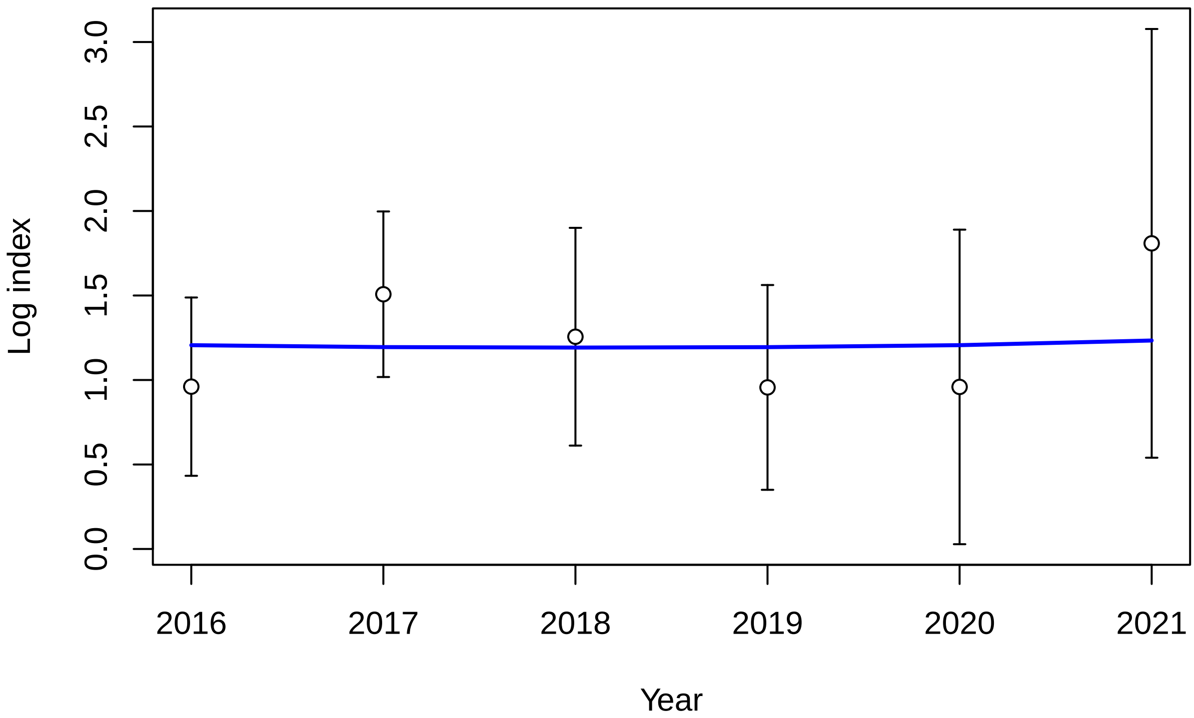
Index
0
5
10
15
20
2016
2017
2018
2019
2020
2021
Year

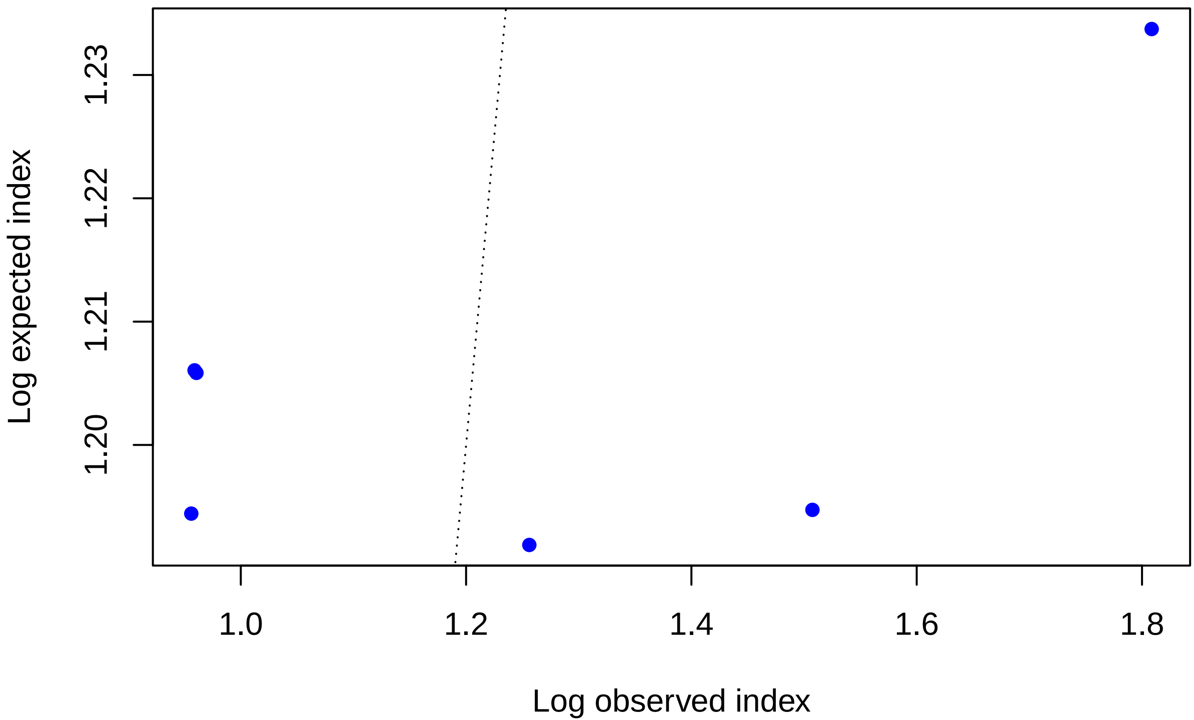


Expected index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
1
2
3
4
5
6
Observed index

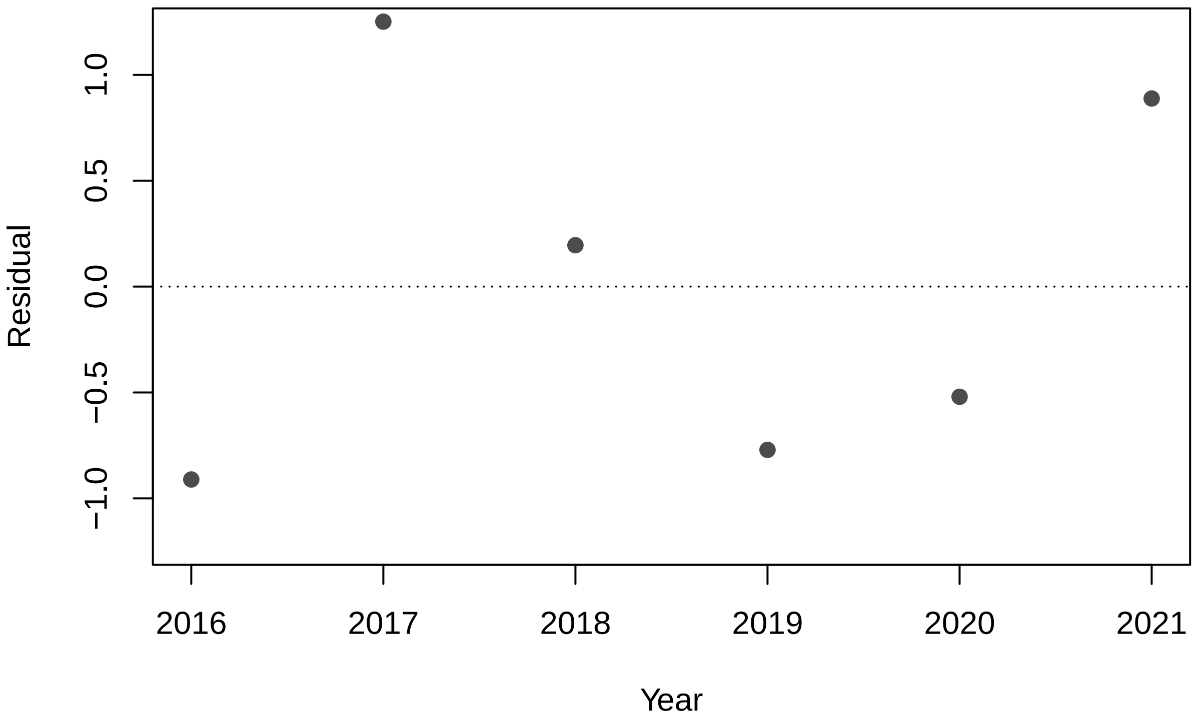


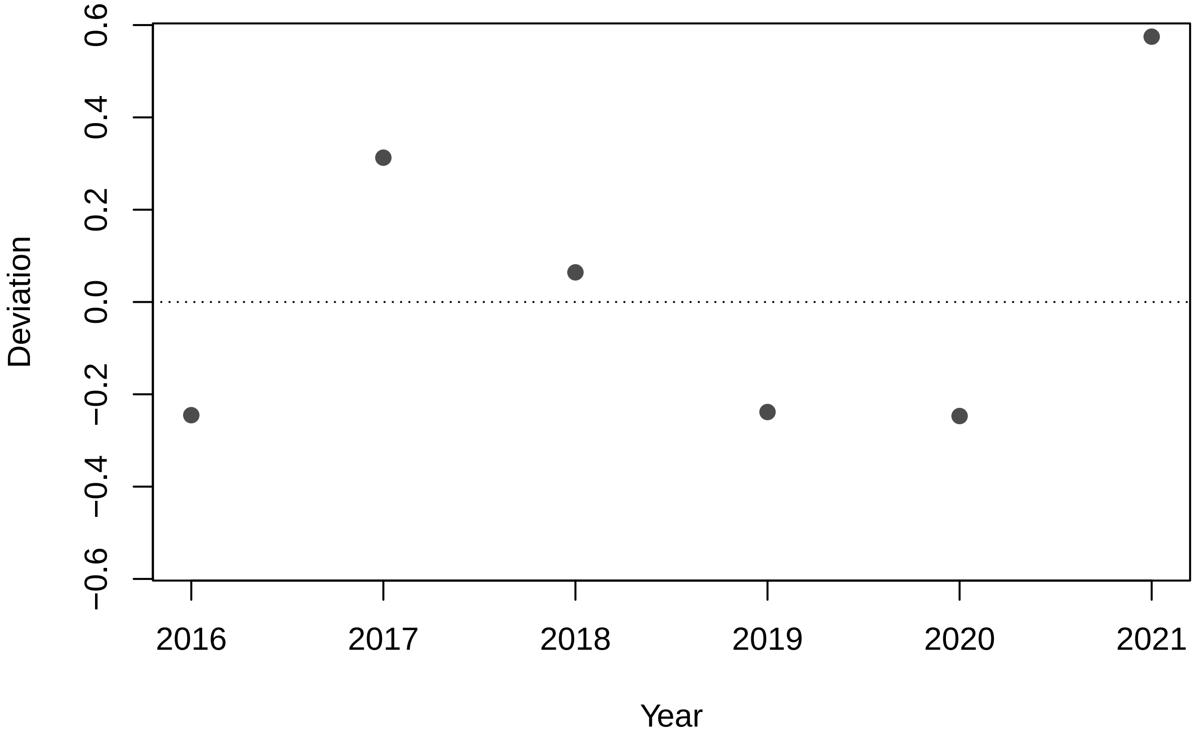
Log index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2016
2017
2018
2019
2020
2021
Year

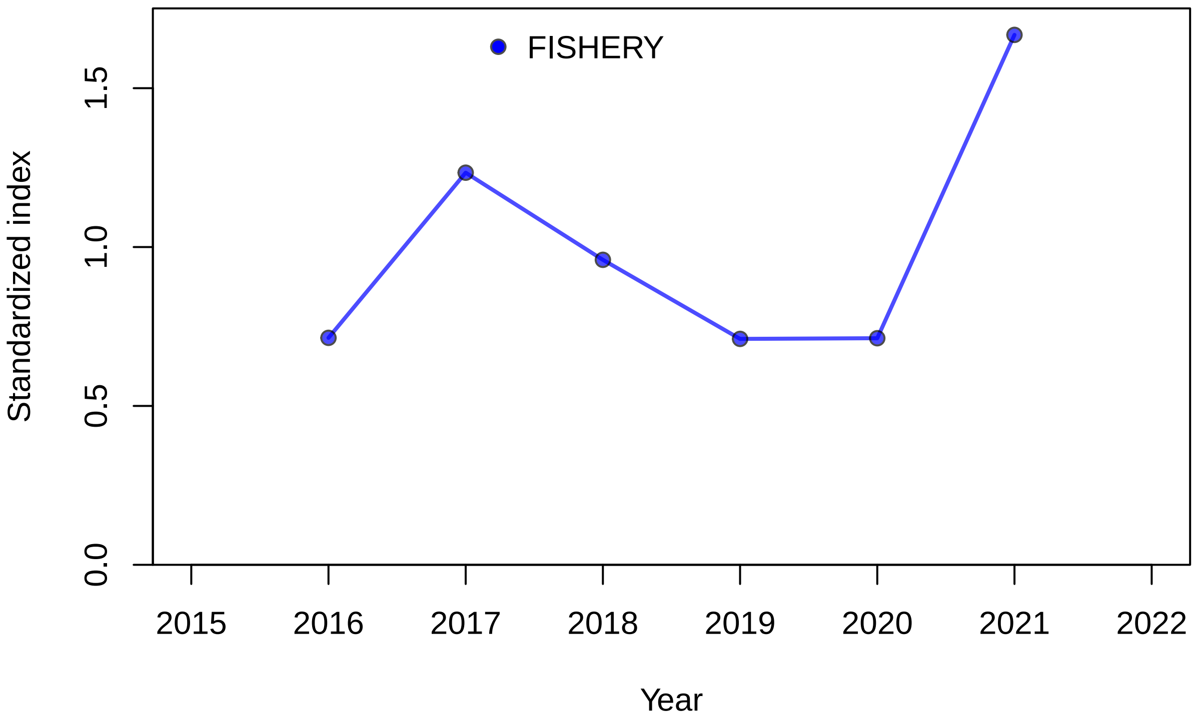

Log index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2016
2017
2018
2019
2020
2021
Year

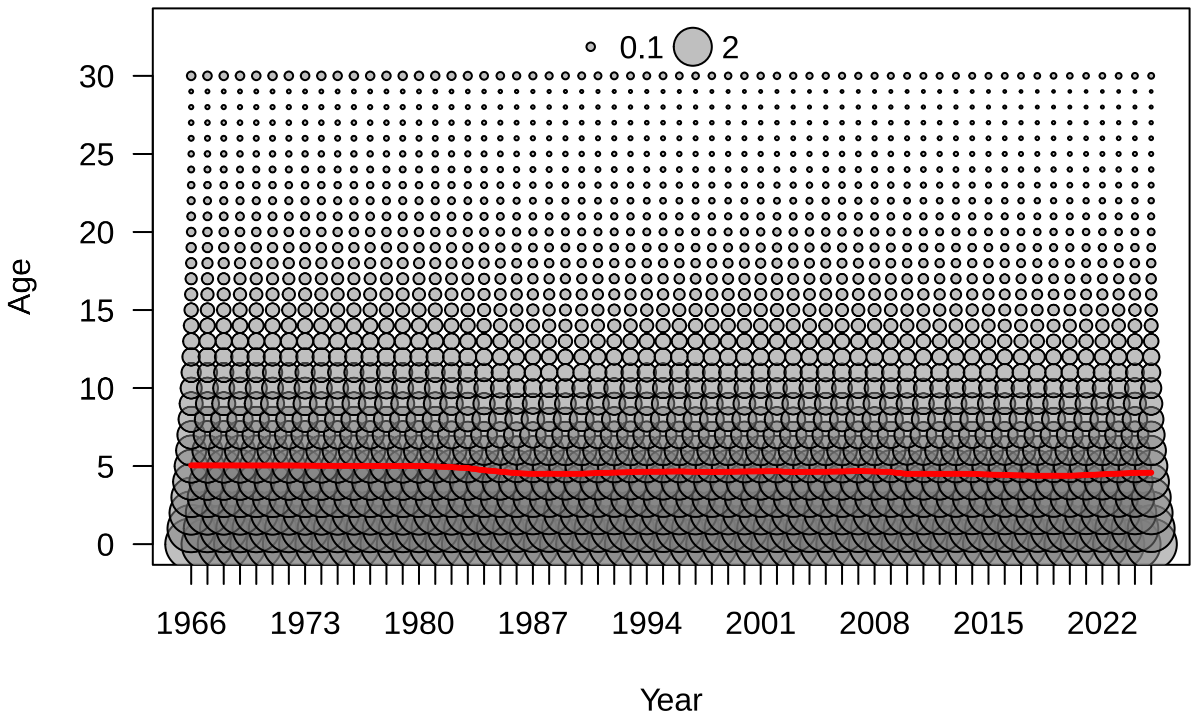


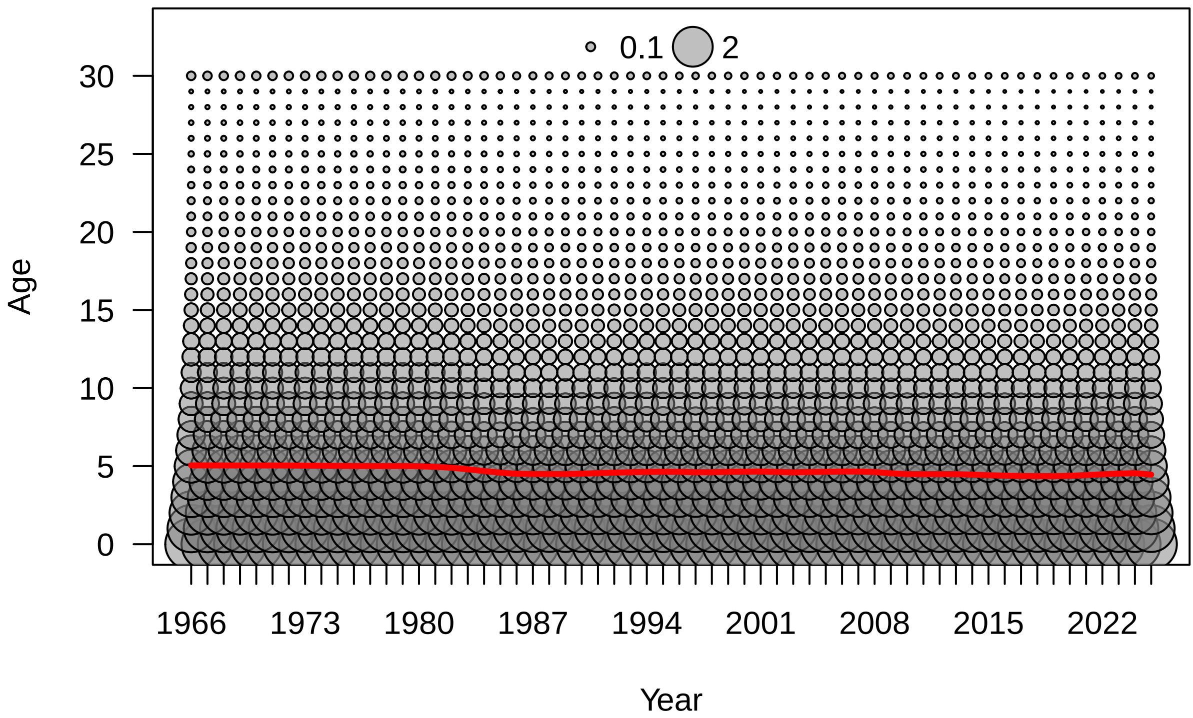
1.20
1.21
1.22
1.23
Log expected index
1.0
1.2
1.4
1.6
1.8
Log observed index

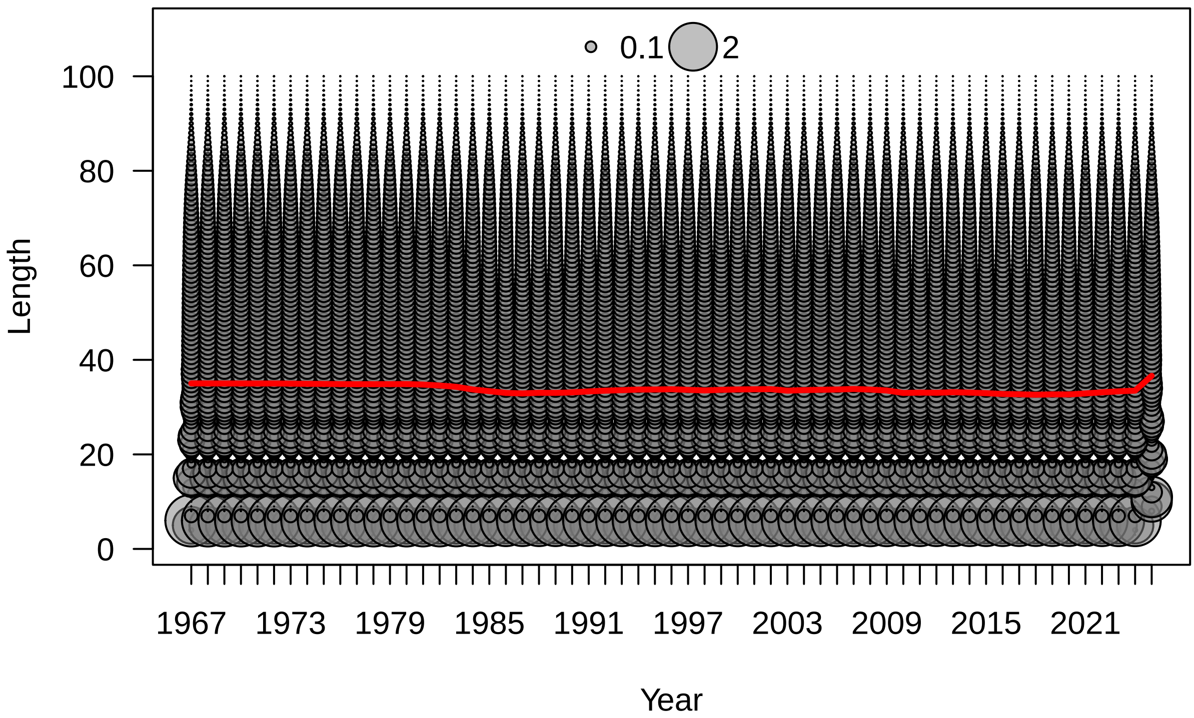

Residual
1.0
0.5
0.0
−0.5
−1.0
2016
2017
2018
2019
2020
2021
Year


Deviation
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
2016
2017
2018
2019
2020
2021
Year

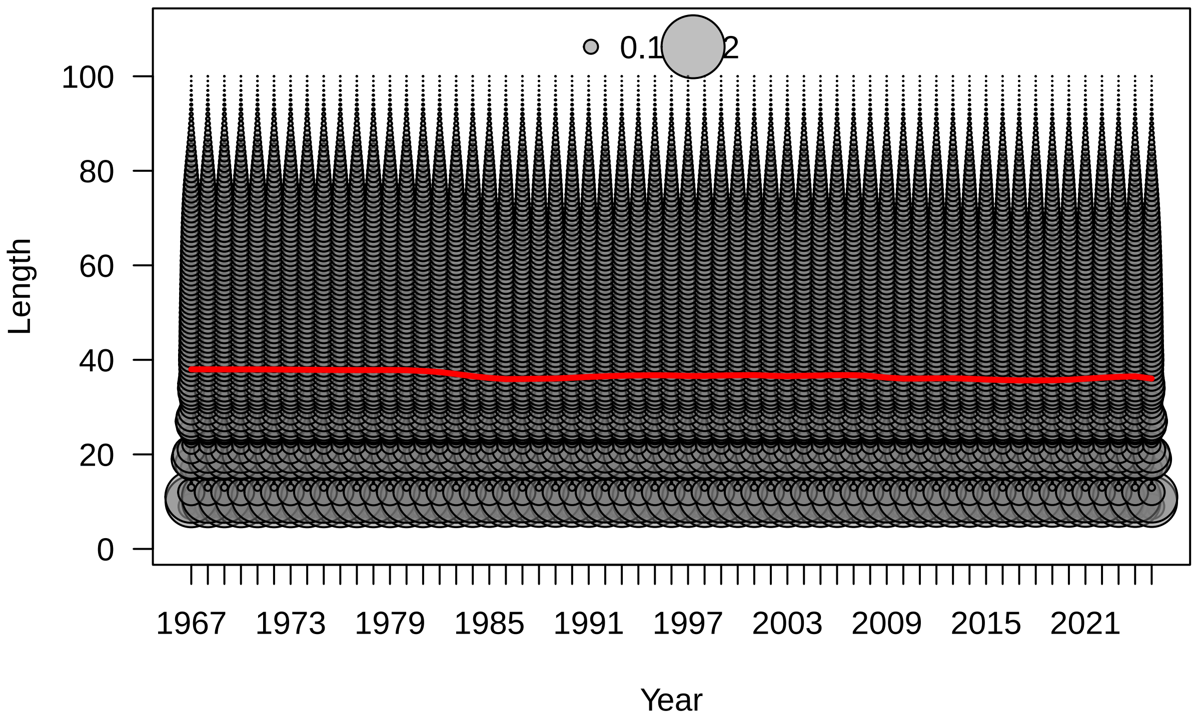

FISHERY
Standardized index
0.0
0.5
1.0
1.5
2015
2016
2017
2018
2019
2020
2021
2022
Year

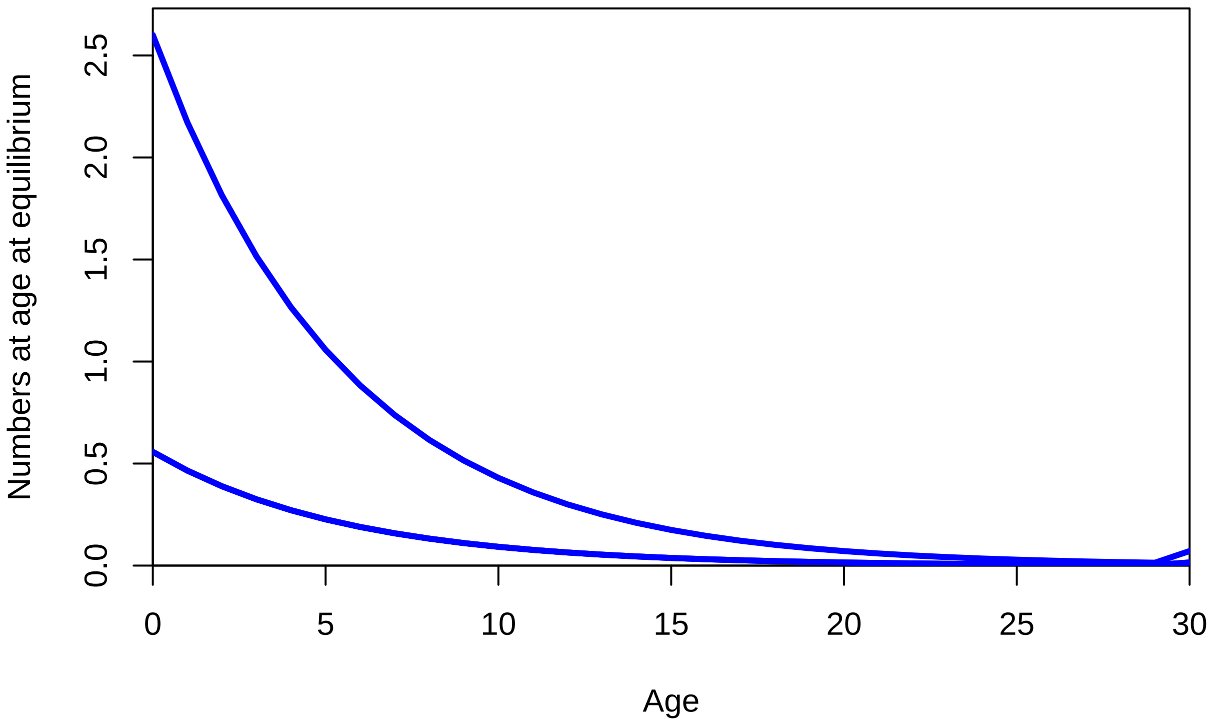

0.1
2
30
25
20
15
10
5
0
Age
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year

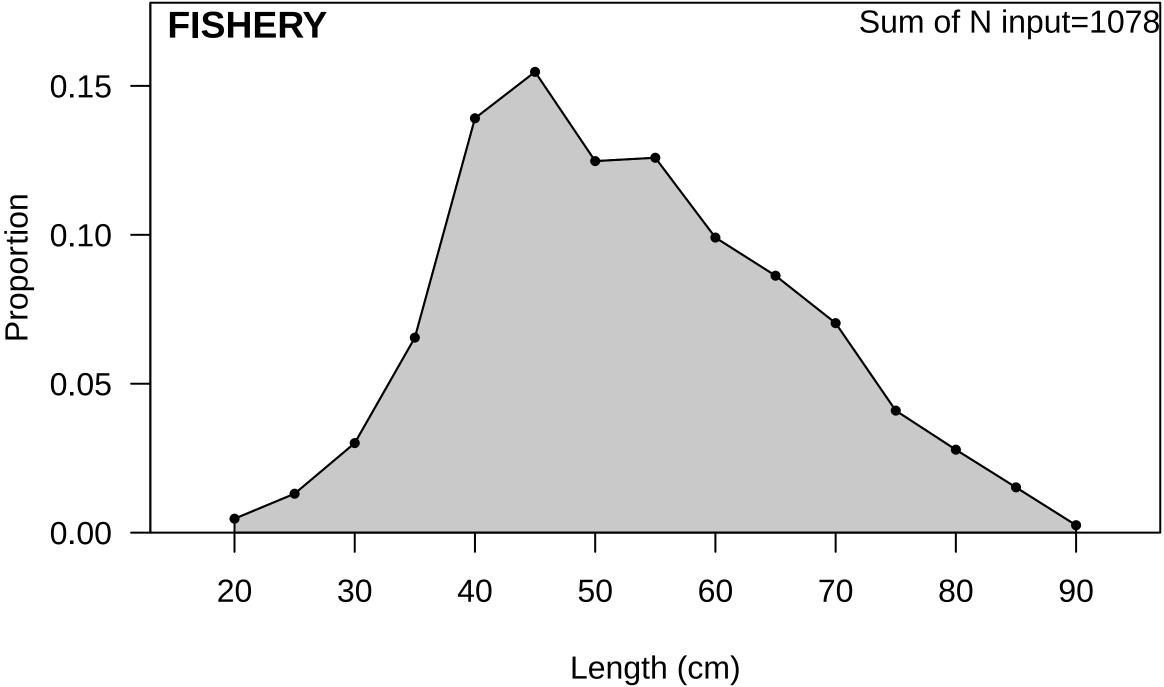

0.1
2
Age
30
25
20
15
10
5
0
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year

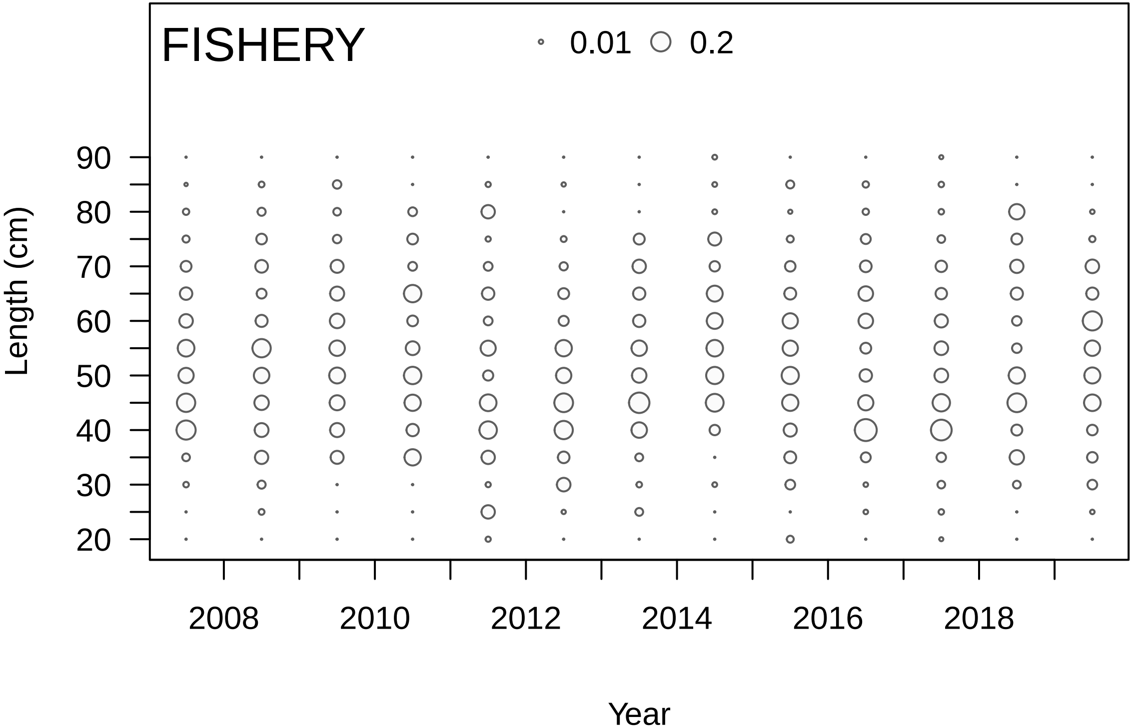


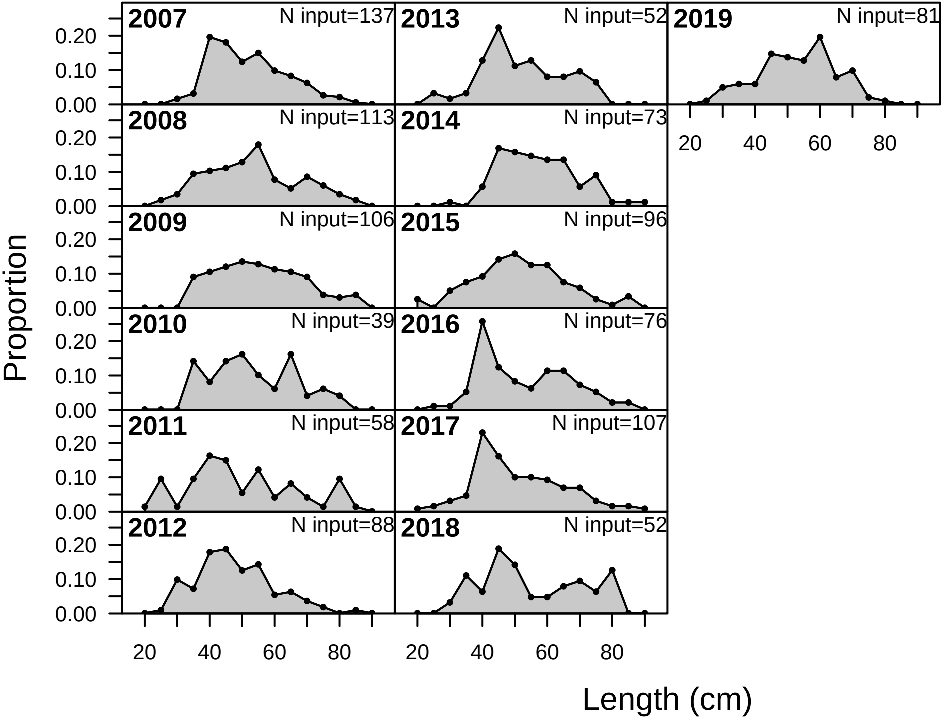
0.1
2
100
80
60
40
20
0
Length
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year


0.1
2
100
80
60
40
20
0
Length
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year

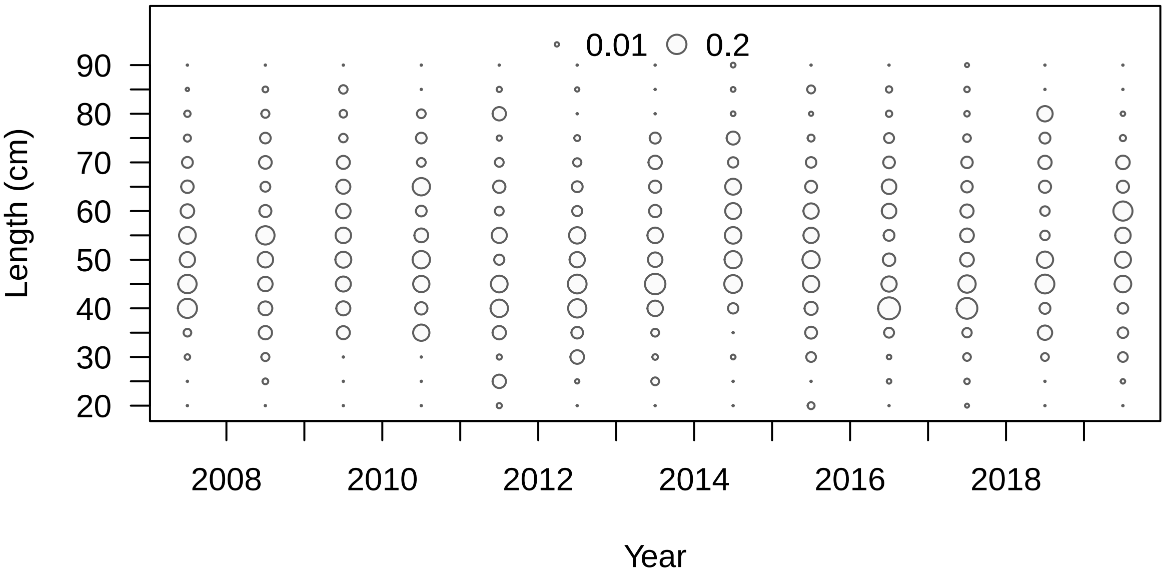

Numbers at age at equilibrium
0.0
0.5
1.0
1.5
2.0
2.5
0
5
10
15
20
25
30
Age

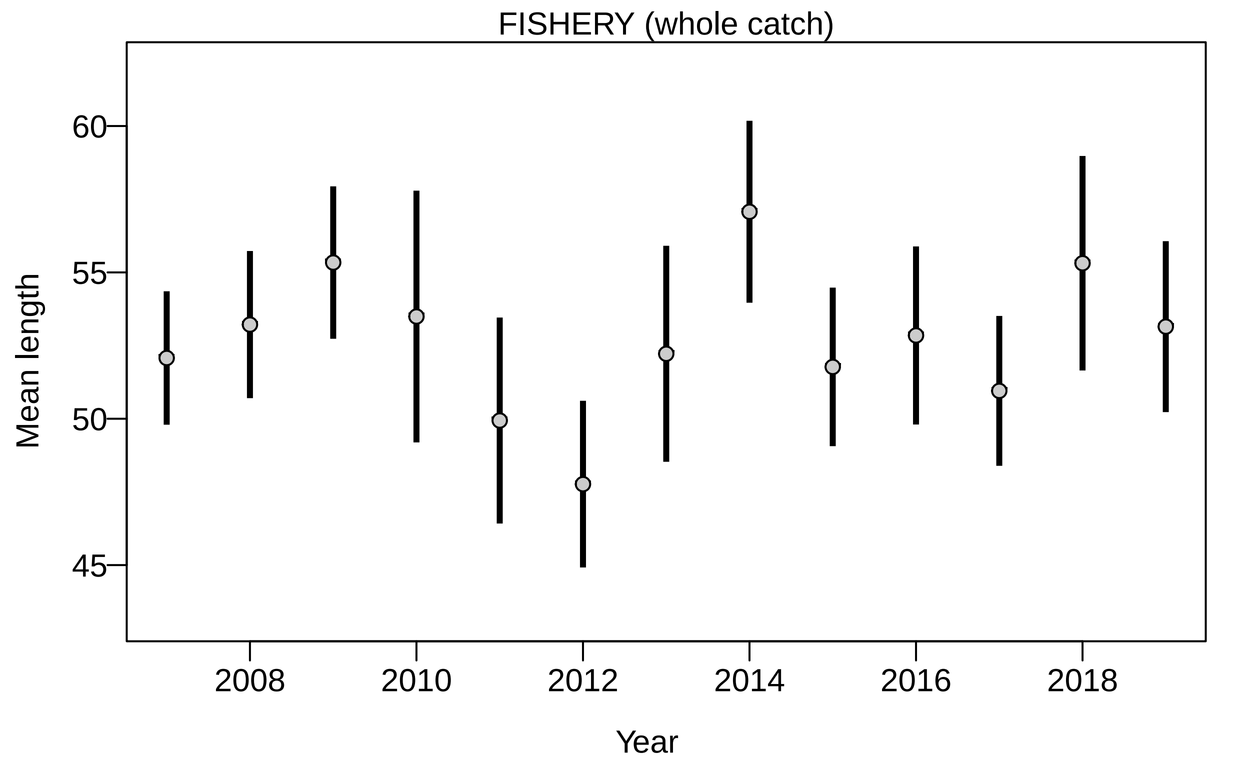

FISHERY
Sum of N input=1078
Proportion
0.00
0.05
0.10
0.15
20
30
40
50
60
70
80
90
Length (cm)

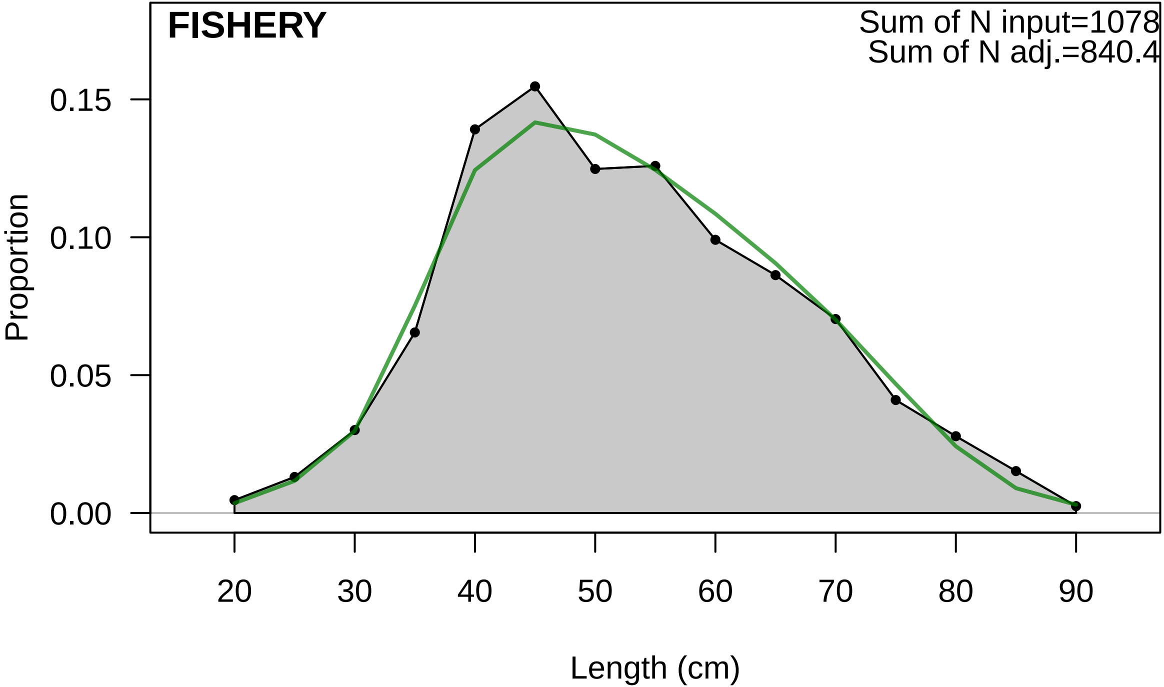

FISHERY
0.01
0.2
Length (cm)
90
80
70
60
50
40
30
20
2008
2010
2012
2014
2016
2018
Year

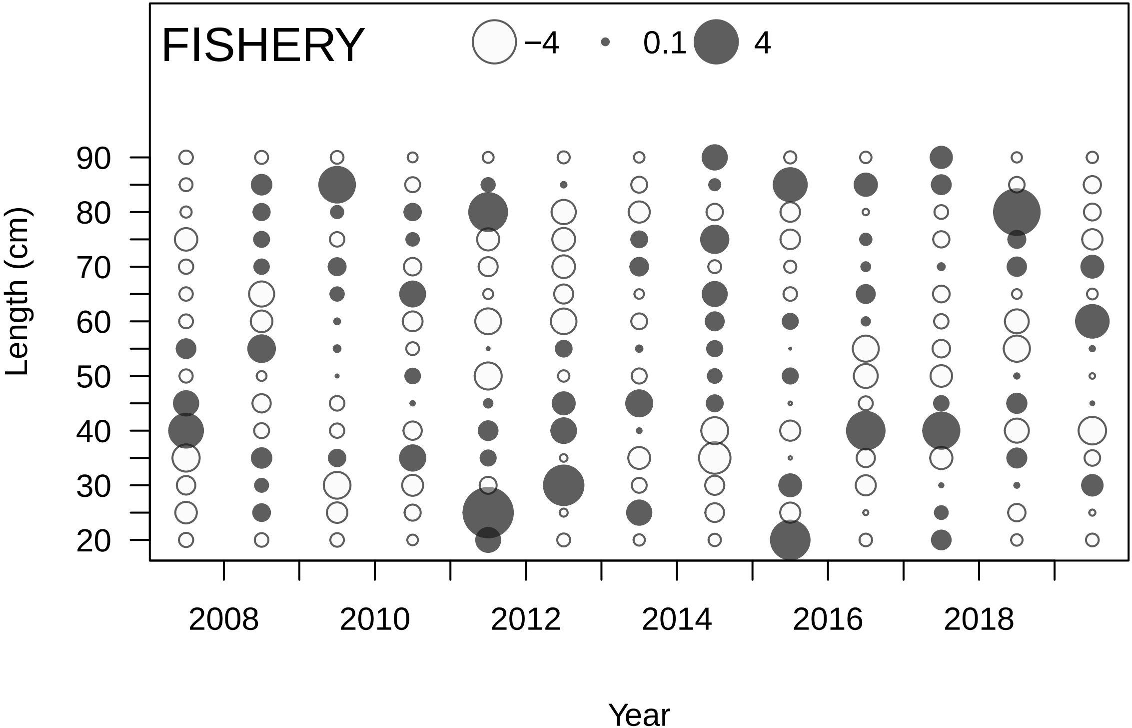


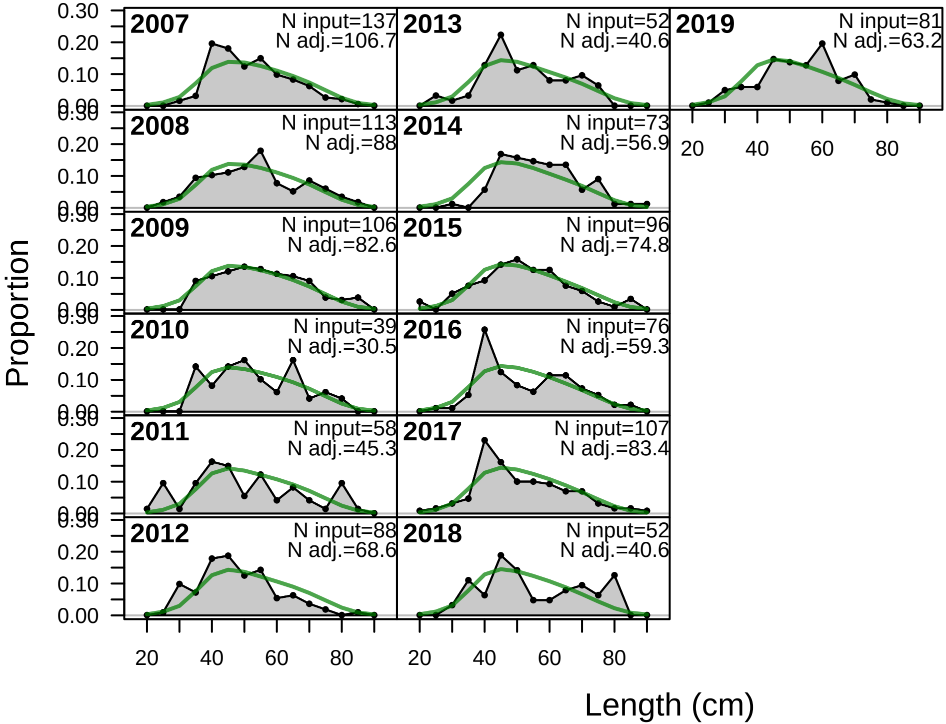
2007
N input=137
2008
N input=113
2009
N input=106
2010
N input=39
2011
N input=58
2012
N input=88
2013
N input=52
2014
N input=73
2015
N input=96
2016
N input=76
2017
N input=107
2018
N input=52
2019
N input=81
0.00
0.10
0.20
20
40
60
80
Proportion
Length (cm)


0.01
0.2
90
80
70
60
50
40
30
20
Length (cm)
2008
2010
2012
2014
2016
2018
Year

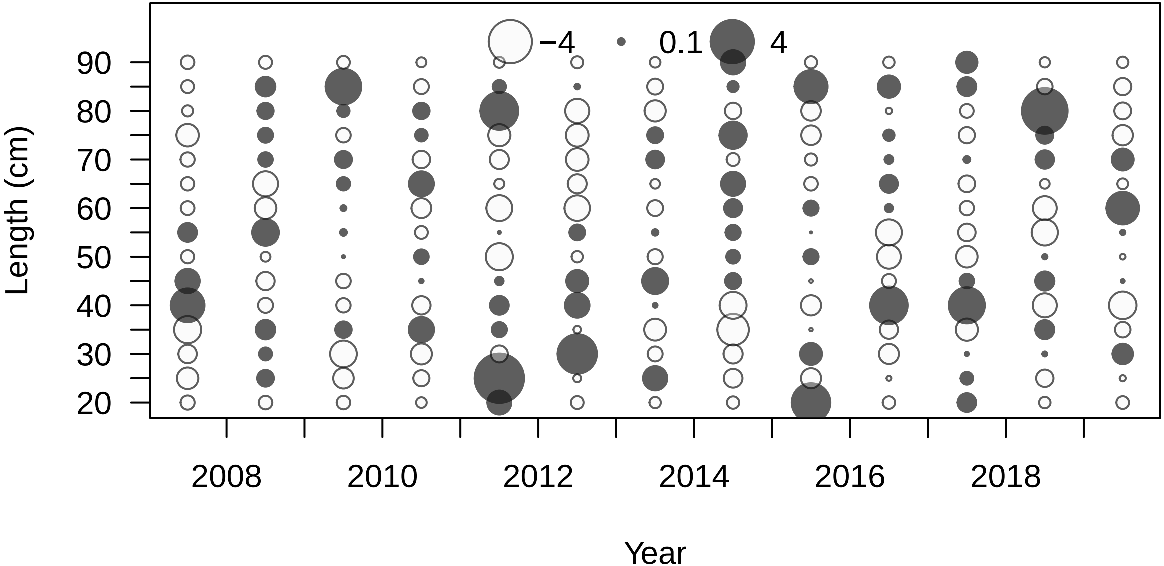

FISHERY (whole catch)
Mean length
60
55
50
45
2008
2010
2012
2014
2016
2018
Year

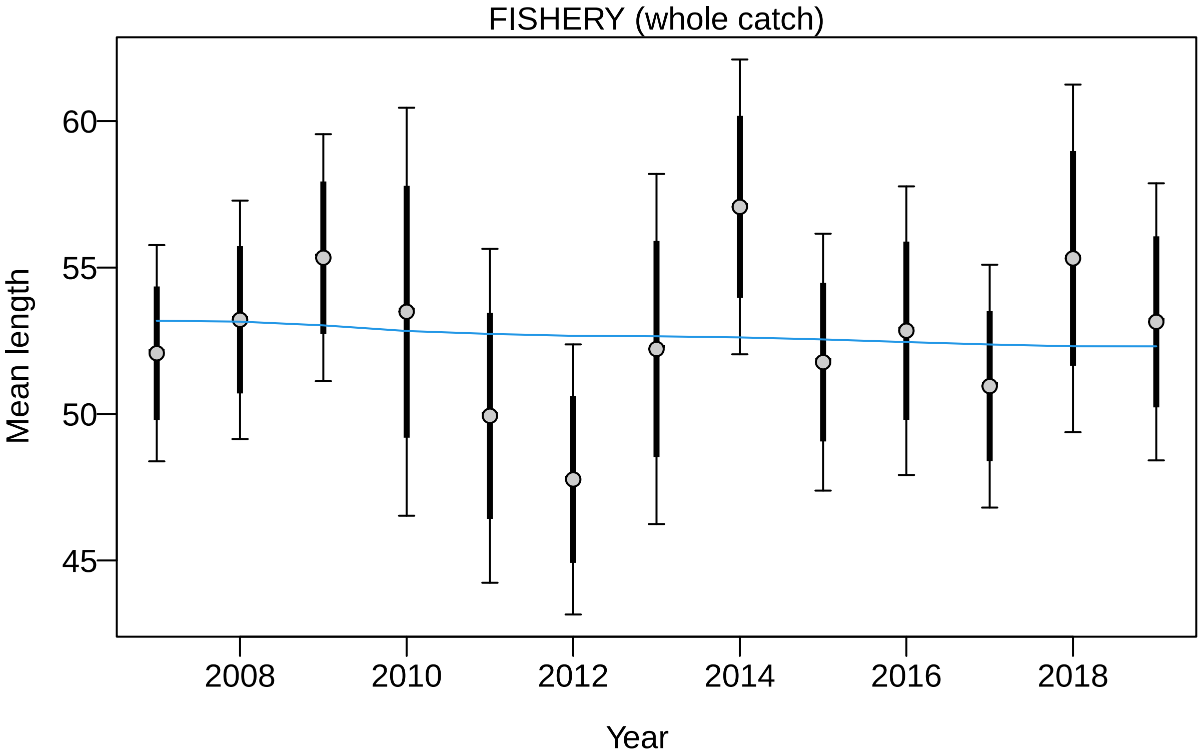

FISHERY
Sum of N input=1078
Sum of N adj.=840.4
Proportion
0.00
0.05
0.10
0.15
20
30
40
50
60
70
80
90
Length (cm)

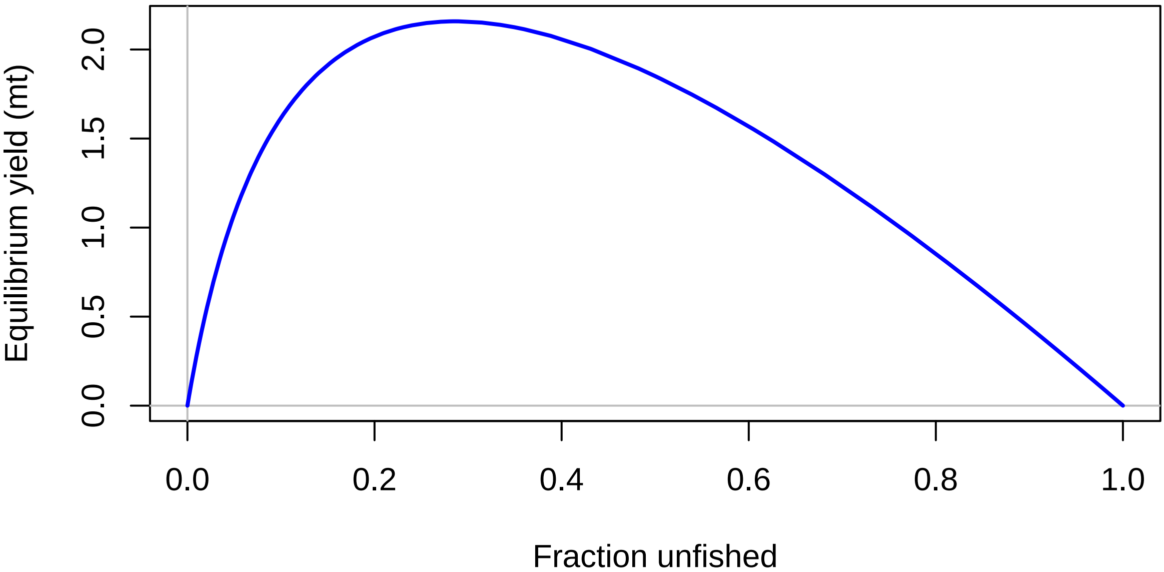

FISHERY
−4
0.1
4
Length (cm)
90
80
70
60
50
40
30
20
2008
2010
2012
2014
2016
2018
Year



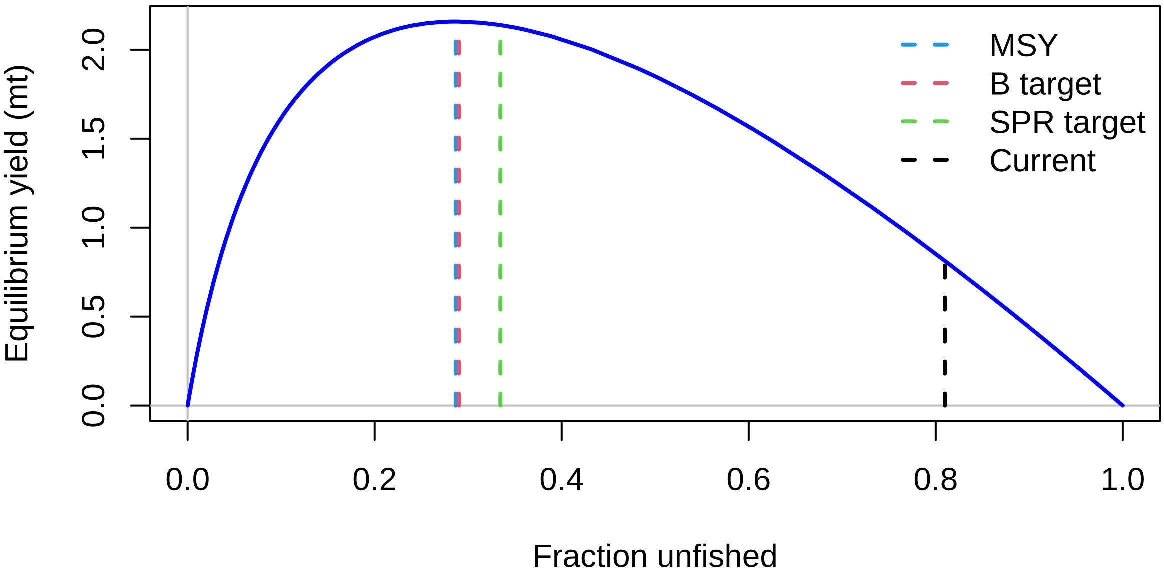
Proportion
Length (cm)
2007
N input=137
N adj.=106.7
2008
N input=113
N adj.=88
2009
N input=106
N adj.=82.6
2010
N input=39
N adj.=30.5
2011
N input=58
N adj.=45.3
2012
N input=88
N adj.=68.6
2013
N input=52
N adj.=40.6
2014
N input=73
N adj.=56.9
2015
N input=96
N adj.=74.8
2016
N input=76
N adj.=59.3
2017
N input=107
N adj.=83.4
2018
N input=52
N adj.=40.6
2019
N input=81
N adj.=63.2
0.00
0.10
0.20
0.30
20
40
60
80

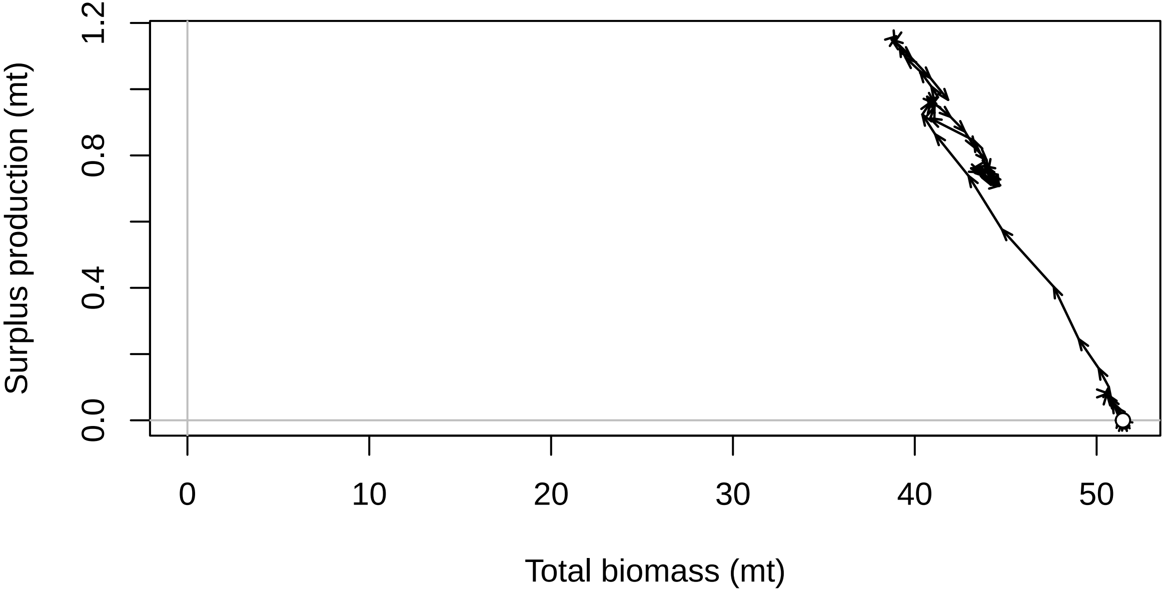

−4
0.1
4
Length (cm)
90
80
70
60
50
40
30
20
2008
2010
2012
2014
2016
2018
Year

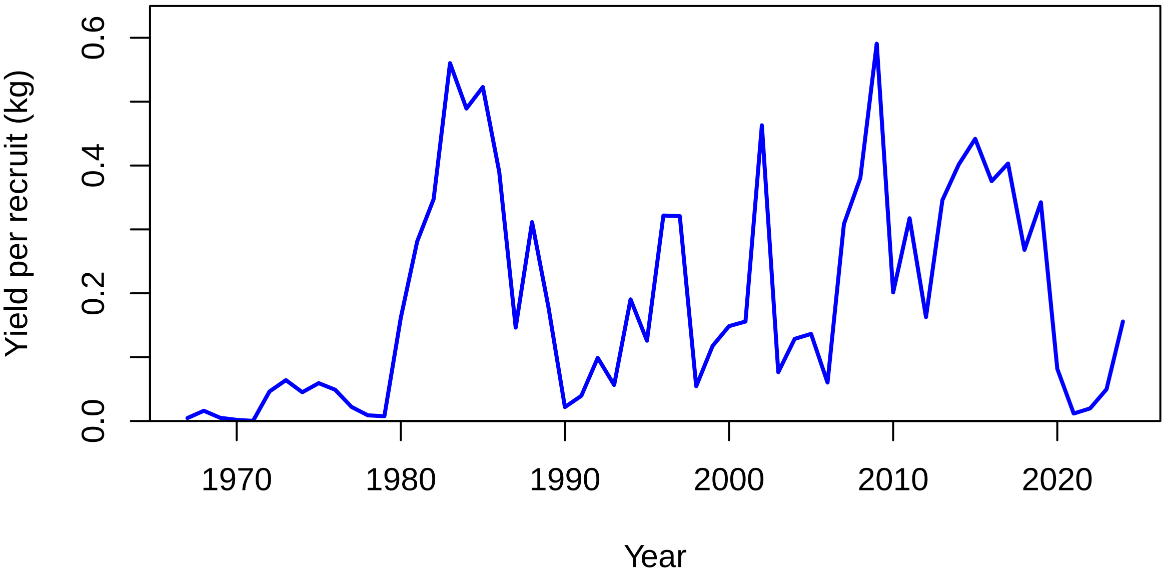

FISHERY (whole catch)
Mean length
60
55
50
45
2008
2010
2012
2014
2016
2018
Year

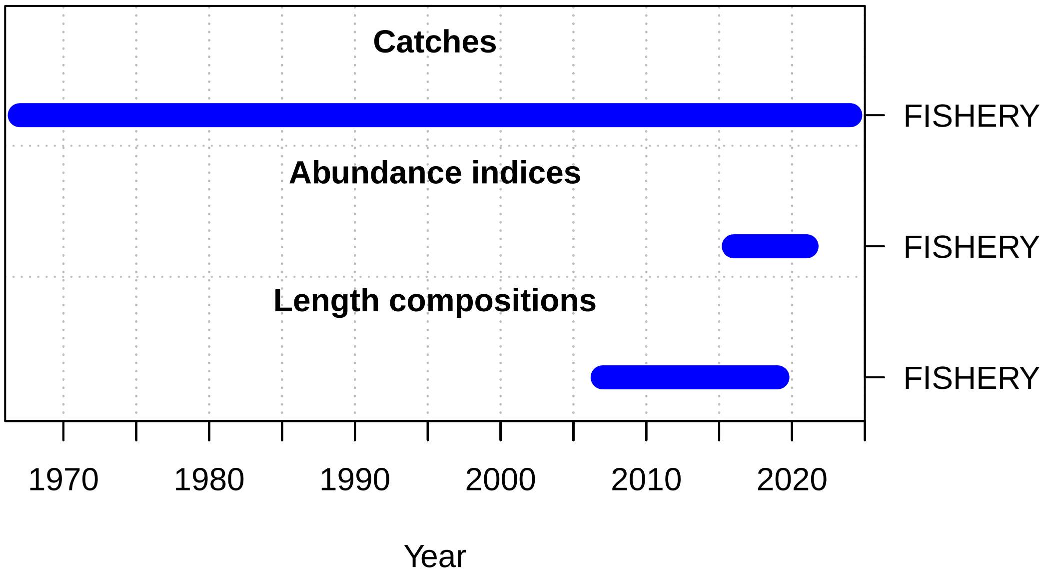


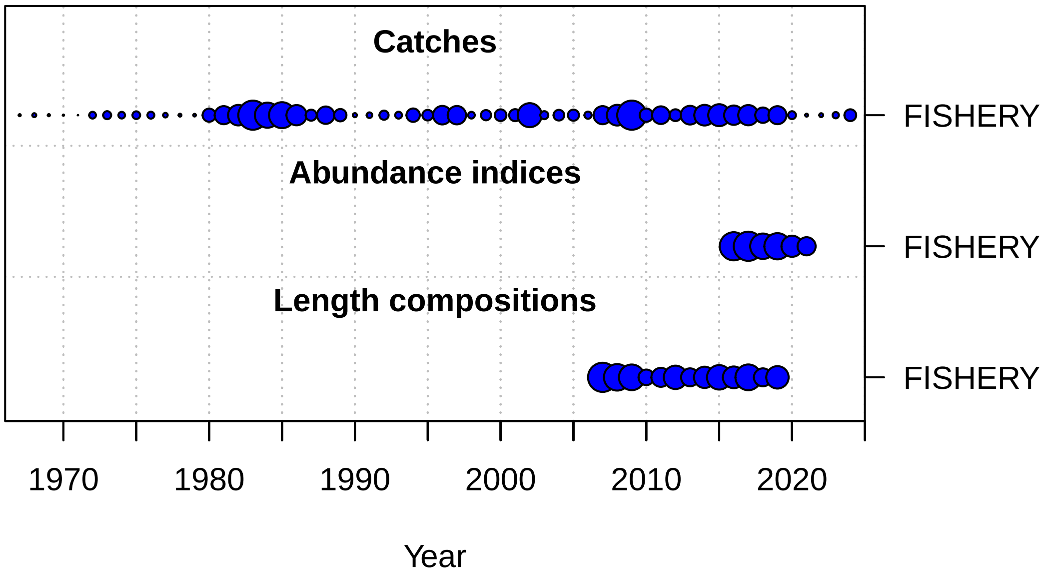
Equilibrium yield (mt)
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Fraction unfished


Equilibrium yield (mt)
0.0
0.5
1.0
1.5
2.0
MSY
B target
SPR target
Current
0.0
0.2
0.4
0.6
0.8
1.0
Fraction unfished

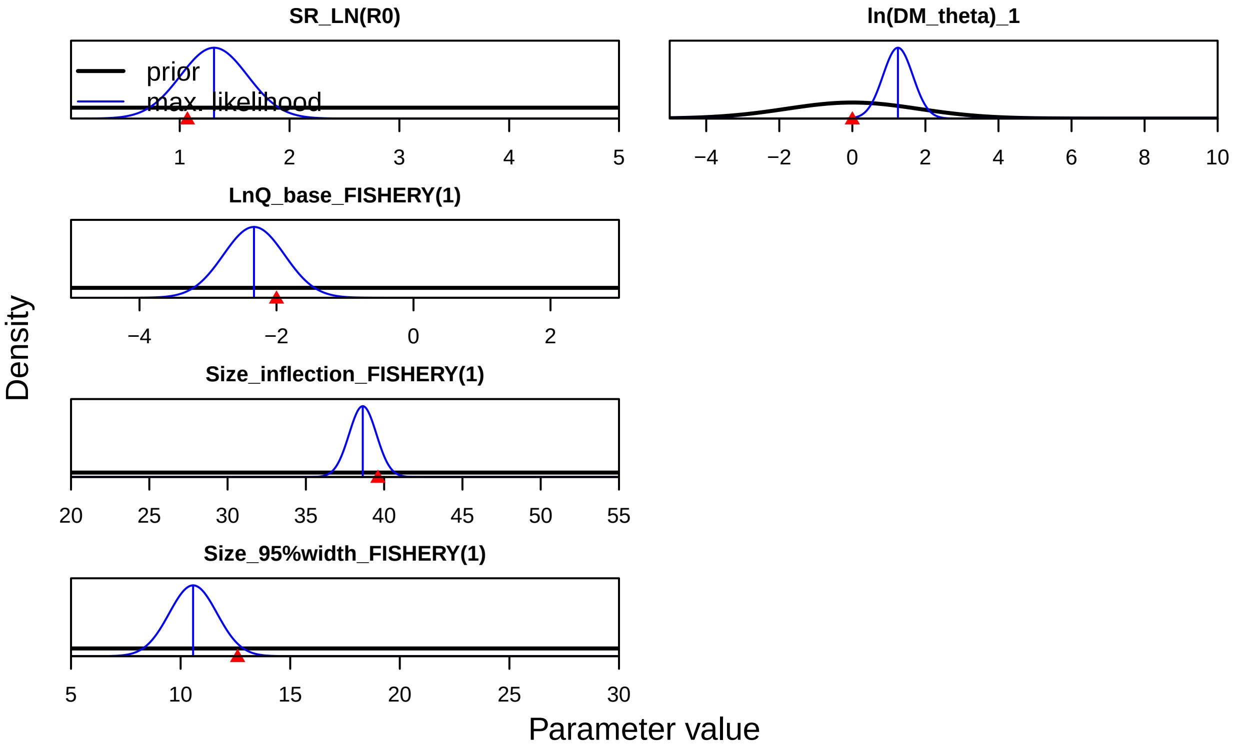

Surplus production (mt)
0.0
0.4
0.8
1.2
0
10
20
30
40
50
Total biomass (mt)


Yield per recruit (kg)
0.0
0.2
0.4
0.6
1970
1980
1990
2000
2010
2020
Year


Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year



Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year



SR_LN(R0)
prior
max. likelihood
1
2
3
4
5
ln(DM_theta)_1
−4
−2
0
2
4
6
8
10
LnQ_base_FISHERY(1)
−4
−2
0
2
Size_inflection_FISHERY(1)
20
25
30
35
40
45
50
55
Size_95%width_FISHERY(1)
5
10
15
20
25
30
Density
Parameter value